



# D&C Specification Brief

## Electrical Services

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25-0437

**Document Issue:**  
B

**Date:**  
4/02/2026  
**Architect:**  
Archipelago

**Project Name:** Skygate Play – Brisbane Airport

**Site Address:** The Circuit, Brisbane Airport



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#### Revision History

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| A     | 17/12/2025 | D&C Pricing Issue         | DE       |
| B     | 04/02/2026 | Revised D&C Pricing Issue | DE       |

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### RELEVANT APPENDICES

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Design for Maintenance Requirements  
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## **1. PRELIMINARIES**

### **1.1 Conditions of Contract**

This section of the specification forms part of the contract documents. Relevant sections of the main contract documents shall be read in conjunction with this part of the specification.

## **2. SCOPE**

### **2.1 General**

The work carried out shall comprise the manufacture, supply, delivery, installation, testing, commissioning and maintenance for the specified period of the Electrical System, as described in detail in the following Clauses and shown on the Drawings.

It is based on the drawings and related documents produced as part of the D&C design phase. As such it reflects the “D&C Tender” stage of documentation. The Electrical Services sub-contractor shall make all allowances necessary to price the project, based on a Design & Construct methodology and style of project delivery.

This document and any associated documents have been produced on the premise that they provide descriptions and information of sufficient detail to enable a full evaluation of the complete project to be gained by:

- Using the descriptions and details provided.
- Relying on the tender’s expertise to make the appropriate allowances by extrapolations, inference, deduction, assumption and the like to ensure a complete and operational mechanical services system is provided that meets and satisfies the clients expectations and operational requirements.
- Where items are not specifically shown on the drawings or included within this document but are required for the type of facility proposed, taking the quality and extent of the provision from the standards set by the information provided in the documentation and including for all such requirements. This document is NOT intended to be an exhaustive brief for the project. It is intended to:
  - Set a minimum standard for the quality both in materials and workmanship for all mechanical services works and systems.
  - Be a basis for the production of documentation to be used to lodge for Building Approval (complete with Form 15 design certification), QFES approvals and For Construction Documentation.

The electrical services drawings and Performance Brief indicate the concept design requirements for this project and set about the scope to allow competitive tenders to be called.

The contactor shall be responsible for the completion of design and detailed services coordination and all design drawings shall be submitted to the project manager by the contractor for final approval, prior to lodgement with all required authorities

All work shall be carried out in accordance with the Specification, Drawings and such details and instructions as are issued by the Superintendent and/or Consulting Engineers during the progress of the work and to the entire satisfaction of the Superintendent.

All work shall be carried out strictly in accordance with manufacturer’s recommendations and industry best practice. The equipment installed shall be handed over to the client in a fully operational state as intended for the equipment function and use.

This Specification shall be read in conjunction with the Drawings, all of which are intended to be mutually explanatory. All work called for by the one even if not by the other, shall be fully

executed. Where ambiguities or discrepancies occur in the documents, the most onerous of them shall be allowed for, except where clarification is sought prior to contract award.

The Drawings and Specification are intended to include everything necessary for the proper completion and setting into effective operation of the installation shown. Minor items as used in normally accepted trade practice with installations of this type and which are not specifically mentioned, shall be included in the Tender and installation.

This Specification relates to workmanship and materials used in the work and shall be applied to equipment supplied or installed by any trade, company, or assisting firm engaged to supply or install any materials and/or equipment.

## **2.2 Fire Engineering Report**

The project is subject to a Fire Engineered solution for a number of issues. The Fire Engineering Report shall be perused to ensure compliance.

## **2.3 Drawing & Design Responsibility**

The Electrical services drawings supplied are diagrammatic and approximate only.

These drawings, together with this performance document and all associated consultant's drawings and briefs are intended to be mutually explanatory and to show the scope of the work required for this D&C Tender Stage. All work set forth by one, if not by the other, shall be fully executed.

The work covered within the drawings and documents produced by the electrical services sub-contractor shall comprise the finalisation of design, co-ordination, documentation, safety in design and construction, supply, installation, testing, commissioning, quality assurance, certification, and maintenance during the defects liability period for the provision of the mechanical services systems as indicated within this document and the following drawings to serve the development.

The electrical services sub-contractor shall be responsible for the detailed design activities listed below in addition to those activities normally undertaken during construction and shall achieve occupational approval for the overall facility. The detailed design shall be fully co-ordinated and compatible with the remainder of the project design.

Design responsibilities include:

- Continue BAC liaison to provide permanent power to site
- Liaise with BAC IT for the incoming fibre and connections to the new project
- Production of client approval drawings.
- Production of co-ordination drawings.
- Production of and submission of drawings for authority approval.
- Production of For Construction drawings.
- Production of As-built drawings.
- Undertake detailed spatial co-ordination of the services with respect to all other works and trades including the detailing of interfaces.
- The contractor is responsible for ensuring all interfaces are covered and that there no scope gaps.
- Perform & confirm calculations for all electrical services requirements

- Design and co-ordinate the supports for the services and equipment.
- Design and select fire stopping for all services penetrations.
- Size and location of penetrations.
- Coordination of interfaces for all trades. Whether or not explicitly identified in the tender documentation, the contractor shall establish all the required interfaces between each service and coordinate the provision of such interfaces. This shall be considered when preparing tender price and undertaking final design.
- Co-ordination of installation with other trades.
- Co-ordination of the construction of the installation.
- Accommodate thermal expansion and construction joint movement in all services taking into account final installation details and consistency with the specified requirements.
- Final selection of all equipment
- The preparation of electrical services drawings sufficient to allow construction, including submissions where required for approvals and authority permits.
- Allow for detailed co-ordination in specific locations eg. Plant rooms, risers, corridors, ceiling & floor voids, etc.
- Technical co-ordination of all the elements of the electrical services works and any suppliers. Where the contractor wishes to use alternative methods or equipment, fully evaluate and submit a detailed report on any proposals for review by the Principles Representative.

## **2.4 Extent of Work**

The work includes but is not limited to the following main elements:

- Design, documentation and certification by an RPEQ Engineer.
- 4 Star Greenstar initiatives (not certification) is being targeted for this building, as such the design and construction of the electrical services shall align with the selected points table as issued by the Greenstar Professional.
- Liaison with Brisbane Airport Corporation for the connection to the power supply to service this building.
- Connection to the proposed BAC transformer
- Conduits, pits and trenching for BAC/NBN Carrier conduits/cables.
- Conduits, pits and trenching for electrical mains to the connection point
- Vacuum excavation to be used around any and all existing services to ensure the existing services are located prior to excavation for any new service.
- Driveway crossing for the electrical mains conduits via under bore methods to BAC approvals. Ensure the access for all vehicles is maintained 24 hours a day. Liaise with BAC if alternate methods to under boring is proposed.
- Reinstatement of all surfaces and compaction as necessary to align as a minimum with the BAC requirements for the different surfaces and areas
- New BAC communication building entry and BAC pits to front boundary.
- Communications and electrical services pits with identification to BAC “pit identification labelling” guidelines.
- Electricity Metering facilities.

- Site Main Switchboard with outgoing connection to building Main distribution switchboard and input connection from the Photovoltaic system.
- Distribution Switchboards.
- Mains and Submains. Main and submains to be sized to include additional power allowance for fitout. Note the fitout loads are in addition to the required spare capacity and are not to use any of the spare capacity.
- Note: All submains shall be sized with a 25% spare capacity in addition to the AS3000 calculated maximum demand. Table C2.
- Determine site maximum demand to AS3000 Table C2 with no further diversification unless approved by BAC.
- Termination of submains at switchboards provided by other trades.
- Electrical and communication services to the 3 phase car chargers
- Supply and installation of the 3 phase car chargers mounted on poles in the car park where determined.
- Part J9D3 Energy metering using smart meters to NCC.
- Metering in accordance with BAC Brisbane Airport Corporation Metering specification.
- BAC Data cabling to kWh meters and the water meters and connection to the BAC metering system.
- Distribution boards (note all to have 25% spare space for future expansion) with NHP Grizz Bar chassis assemblies on all switchboards. We note the 25% spare space is at practical completion so the contractor must make allowance for the tenants fit-out circuits in addition to the 25%.
- Distribution boards shall also be sized to incorporate Zencontrol lighting control systems and Rapid RCD testing facilities.
- Establishment of a BAC net building controller network for the Building Management System.
- BAC communication cabinet, switches, cable management, fibre and data cabling network.
- All external lighting and security lighting to be designed for compliance with AS1158, AS4282, BAC and CASA standards.
- Lighting to handstand area and awnings.
- Security lighting to the full perimeter of the building.
- Lighting and power to the internal areas and amenities.
- Power to equipment
- IP66 minimum vandal proof fittings over all external doors with IP66 sensor. 20 lux minimum to be achieved at ground level 3 metres from the door.
- Lighting control system to control the lighting to all areas. Zencontrol Dali 2, control systems shall be used as per the airport recent Technical Note TN-58. Use a combination of motion sensors, time clocks, PE cells and manual switching. Provide contactors on all lighting circuits to allow connection to the tenants alarm system.
- Motion sensors to lights in amenities, stores, plantrooms, etc.
- A PE cell initiated control for the external lighting with a minimum 4ch – time switch to switch lights off at curfew hour.
- Time switch and PE cell controlled power supplies to external signs.



- Relocation of any existing streetlighting that clashes with new driveway crossovers, additional streetlighting maybe required if the relocation is not sufficient to maintain the rad lighting category to AS1158
- Earthing.
- Cabling and cable support systems.
- Trenching, conduits and pits for underground cabling systems.
- Accessories and appliances.
- General and specific power systems.
- Equipment supports and bracing.
- Luminaires.
- Lighting Control Systems.
- Emergency Evacuation Lighting.
- Lightning protection assessment and installation if required
- Conduit and draw wire installation for any Electronic Security to gates and doors
- Conduit and draw wire installation for a CCTV system.
- Conduit and draw wire installation for any tenant special community or security services within the tenancies
- Coordinate any inground conduits with all other services and structure.
- Underground utilities locating.
- Liaison with other trades.
- All design drafting to be performed in Revit Model including as built drawings. Refer to BAC drafting standards for further information.
- Testing and commissioning
- Maintenance Manuals.
- As-Installed Drawings.
- Training.
- Maintenance.

The design and installation shall take all of the standards and reference documents as issued by Brisbane Airport Corporation, including the tenant specific brief for this project.

## **2.5 Work In Other Trades**

Related work in connection with the Electrical Services (performed by other trades) includes the following:

### **2.5.1 By Main Contractor/Builder**

Coordination of all trades.

Site facilities and ablutions.

Disposal of spoil & rubbish from site.

Storage space for tools and materials on site.

Site temporary water and electrical services including site lighting.

Fixed elevated work platforms, scaffolding, edge protection and the like.

Lifting equipment such as crane, hoist and personnel lift.

Equipment cupboards.

Signage to equipment cupboards including switchboard cupboards or rooms, communications cupboards or room, fire panel cupboards or fire control rooms.

Core holes including x-raying.

Coordinating set outs for electrical services including surveying.

Provision of equipment requiring electrical connection as detailed.

Co-ordination with Electrical Services to allow installation of concealed conduits and wiring.

## 2.5.2 By Mechanical Services

Provision of mechanical services equipment.

Provision of mechanical services switchboard.

Power and control wiring to mechanical equipment from the MSSB.

Connection to field equipment from isolator provided by Electrical Services

## 2.5.3 By Hydraulic Services

Provision of hydraulic services equipment and control panels

## 2.5.4 By Fire Protection Services

Provision of Automatic Fire Detection and Alarm System, Automatic Sprinkler System and Emergency Warning System.

Provision of 4G/GSM cards for Fire Brigade connection

## 2.6 Design Deliverables

| Project Phase    | Milestone | Description of Documentation Required                                                                                                                                                                                                                                                                                                                 |
|------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Schematic        | 15%       | <p>Overall system topology identifying incoming supply connections and major infrastructure (substations, generators, UPS, main switchboards, distribution boards)</p> <p>Major infrastructure sizing (eg maximum demand, power ratings, backup durations)</p> <p>Lighting protection risk assessment including recommended mitigation strategies</p> |
| Developed Design | 40%       | <p>Update Schematic Design documentation</p> <p>Cable routes identified</p> <p>Concepts for control systems identified (eg load shedding or other power flow control philosophies)</p>                                                                                                                                                                |
| Detailed Design  | 90%       | <p>Update Developed Design documentation</p> <p>Major infrastructure sizing confirmed</p> <p>Cable routes confirmed and sized</p> <p>Control system philosophies confirmed</p>                                                                                                                                                                        |

|                   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Construction      | 100% | <p>Updated Detailed Design</p> <p>Design certificates confirming all works have been designed in accordance with BAC standards and requirements</p> <p>Calculation summary report identifying:</p> <ul style="list-style-type: none"> <li>• Maximum demand at each switchboard (Amps)</li> <li>• Voltage drop at each switchboard (%)</li> <li>• Prospective fault rating at each switchboard (kA)</li> <li>• Circuit protection discrimination</li> </ul>                                            |
| Post-Construction |      | <p>As built drawings including:</p> <ul style="list-style-type: none"> <li>• Installation certificates confirming all works have been installed in accordance with BAC standards and requirements</li> <li>• Updates to BAC main switchboard and building wide low voltage reticulation schematics</li> <li>• Switchboard general arrangement, details and schematics</li> <li>• All testing results</li> <li>• All independent inspection reports (especially relevant for 11 kV systems)</li> </ul> |

## 2.7 BAC Current Suppliers

"LED" Lighting Suppliers (Internal).

- Minimum 10 year warranty.
- Advanced Lighting Technologies – Scott McCarthy 0474 510 471.
- LAD - Roland Poon 0406 906 779.
- Gerard Lighting -
- Raylinc - Gary Watson 0408 221 585.
- Zumtobel – David Dehan 0409 472 841.

"LED" Lighting Suppliers (External).

- Minimum 10 year warranty.
- Advanced Lighting Technologies - Scott McCarthy 0474 510 471.
- LAD - Roland Poon 0406 906 779.
- Raylinc - Gary Watson 0408 221 585.

"LED" Exit and Emergency Lighting Suppliers.

- Clevertronics – Hive- Control.

Street Lighting Works.

- Advanced Lighting Technologies - Scott McCarthy 0474 510 471.

"LED" External Carpark and Hardstand Lighting.

- Light poles to be GM poles, tapered, galvanised finish, access panel (or approved by BAC).
- LED lights to be from above suppliers (or approved by BAC).

MSB and DB Manufacturers

- Switchboard Innovations.
- Sunline.
- Trivantage Switchboards.

- NHP.
- Schneider.
- Lai Switchboards.
- Reids Switchboards.
- J&P Richardson

### 3. GENERAL REQUIREMENTS

#### 3.1 General

This section outlines general requirements which are applicable to all sections of this specification and the works.

#### 3.2 General Compliance with Regulations

The work carried out by the Contractor shall comply in all respects with this brief and The National Construction Code – Volume One – BCA.

BAC (Brisbane Airport Corporation) standards and design guidelines - referenced in detail below. Supply Authorities.

The relevant Acts governing installation of the contract works in this class of building.

The requirements of the Chief Inspectors of the relevant Statutory Authorities.

ACMA Regulations.

ARFF (Airport Fire).

Relevant Australian Standards.

Any other regulations that apply directly or indirectly to such installations in the locations.

Where conflict arises between this brief and any of the applicable Acts, Codes or Standards, the highest standard of materials and workmanship shall prevail.

#### 3.3 BAC Reference Documents and Industry Standards, Regulations and Guidelines

| Reference | Title                                                    |
|-----------|----------------------------------------------------------|
| AAG-601   | Electricity Applications                                 |
| AAG-602   | Embedded Generation Applications                         |
| ACG-601   | Arc Flash Procedure                                      |
| ACG-602   | Embedded Generation Procedure                            |
| ACG-701   | Communications Infrastructure Installation Specification |
| ADG-020   | Sustainable Design                                       |
| ADG-525   | Skygate Tenancy Design                                   |
| ADG-601   | Electrical                                               |
| ADG-602   | Embedded Generation                                      |
| ADG-605   | Lighting                                                 |
| ADG-635   | Metering                                                 |
| ADG-701   | Communications Infrastructure Design Specification       |
| ADG-525   | Skygate Tenancy Design                                   |
| ACG-701   | Communications Infrastructure Installation Specification |

|         |                                          |
|---------|------------------------------------------|
| ADG-903 | BIM and CAD Standards                    |
| ADG-001 | Introduction and Overview                |
| ADG-020 | Sustainable Design                       |
| ADG-010 | Design Principles                        |
| ADG-015 | Safety in Design                         |
| ADG-050 | Project Completion Documents             |
| ADG-051 | Commissioning                            |
| ADG-625 | Building Management System (BMS)         |
| TN-52   | Rapid Test in electrical Switchboards    |
| TN-58   | Zencontrol Dali2 lighting control system |
|         |                                          |
|         |                                          |
|         |                                          |

## INDUSTRY STANDARDS, REGULATIONS AND GUIDELINES

| Reference        | Title                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AS/NZS 1158      | Lighting for roads and public spaces                                                                                                                                                   |
| AS 1428          | Design for access and mobility                                                                                                                                                         |
| AS/NZS 1680      | Interior lighting                                                                                                                                                                      |
| AS 1798          | Lighting poles and bracket arms – Recommended dimensions                                                                                                                               |
| AS/NZS 3000      | Electrical installations (known as the Australian/New Zealand Wiring Rules)                                                                                                            |
| AS/NZS 3008.1.1  | Electrical installations - Selection of cables – Cables for alternating voltages up to and including 0.6/1 kV – Typical Australian installation conditions                             |
| AS 4282          | Control of the obtrusive effects of outdoor lighting                                                                                                                                   |
| AS/NZS 4665.2    | Performance of external power supplies – Minimum energy performance standard (MEPS) requirements                                                                                       |
| AS/NZS 4782      | Double-capped fluorescent lamps                                                                                                                                                        |
| AS/NZS 4783      | Performance of electrical lighting equipment                                                                                                                                           |
| AS/NZS 4847      | Self-ballasted lamps for general lighting services                                                                                                                                     |
| AS/NZS 4879      | Performance of transformers and electronic step-down convertors for ELV lamps                                                                                                          |
| AS/NZS 5110      | Recessed luminaire barriers                                                                                                                                                            |
| AS 60529         | Degrees of protection provided by enclosures (IP Code)                                                                                                                                 |
| AS/NZS 60598     | Luminaires                                                                                                                                                                             |
| AS/NZS 60923     | Auxiliaries for lamps                                                                                                                                                                  |
| AS/NZS 61000     | Electromagnetic compatibility (EMC)                                                                                                                                                    |
| AS/NZS 61347     | Lamp control gear                                                                                                                                                                      |
| AS/NZS IEC 62560 | Self-ballasted LED-lamps for general lighting services by voltage > 50 V – Safety specifications                                                                                       |
| ANSI C78.377     | Specifications for the Chromaticity of Solid State Lighting Products                                                                                                                   |
| ANSI C136.10     | American National Standard for Roadway and Area Lighting Equipment - Locking-Type Photocontrol Devices and Mating Receptacles - Physical and Electrical Interchangeability and Testing |
| ANSI C136.31     | American National Standard for Roadway and Area Lighting Equipment – Luminaire Vibration                                                                                               |
| IEC 60188        | High-pressure mercury vapour lamps                                                                                                                                                     |

|                                                                                         |                                                                          |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| IEC 60662                                                                               | High-pressure sodium vapour lamps                                        |
| IEC 61167                                                                               | Metal halide lamps                                                       |
| IEC 62031                                                                               | LED modules for general lighting                                         |
| IES LM-79                                                                               | Electrical and Photometric Measurements of Solid-State Lighting Products |
| IES RP-8                                                                                | Roadway Lighting                                                         |
| IES TM-21                                                                               | Projecting Long Term Lumen Maintenance of LED Light Sources              |
| IES TM-30                                                                               | Method for Evaluating Light Source Color Rendition                       |
| Civil Aviation Safety Authority Manual of Standards (MOS) Part 139 – Aerodromes         |                                                                          |
| Queensland Department of Transport and Main Roads Specifications – MRTS94 Road Lighting |                                                                          |

### 3.4 Hold Points during Construction

Inspections of the installation during construction by BAC and the peer review consultants are required at the following stages:

First rough in of cabling prior to the ceiling and wall sheeting installation.

Underground conduit installation, prior to backfilling of trench.

Completion and commissioning stage.

### 3.5 Functional Specification

#### Design Life:

| Element                                                 | Minimum Design Life                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------|
| Batteries                                               | 10 years                                                                               |
| Cable containment – Above ground (trays, conduits, etc) | 25 years                                                                               |
| Cable containment – Below ground (conduits, pits, etc)  | 50 years                                                                               |
| Electricity generators                                  | 25 years                                                                               |
| Ground power units                                      | 10 years                                                                               |
| High voltage switchgear                                 | 10,000 operations (normal conditions)<br>30 operations (at full short-circuit current) |
| High voltage systems                                    | 25 years generally<br>40 years for cable                                               |
| Lightning protection                                    | 25 years                                                                               |
| Low voltage and extra-low voltage cables                | 25 years                                                                               |
| Low voltage contactors                                  | 10 million operations                                                                  |
| Low voltage switchboards – Indoors                      | 25 years                                                                               |
| Low voltage switchboards – Outdoors                     | 15 years                                                                               |
| Programmable logic controllers                          | 15 years                                                                               |
| Uninterruptible power supplies                          | 10 years                                                                               |

### 3.6 Quality of Materials and Workmanship

All materials shall be new, of the best quality and of the class most suitable for the purpose specified.

All equipment shall have a verified history of successful use in the commercial environment. No prototype equipment will be accepted.

All work on the site shall be supervised by a competent tradesman supervisor who shall be present on site when requested by the Superintendent, for all phases of the project. A representative shall be nominated, who shall be present and responsible for the day to day running of the project.

The entire electrical installation shall be executed by licensed, experienced tradesmen, and shall be completed in a neat and first-class tradesman-like manner to the satisfaction of the Architect.

### **3.7 Product Ordering & Lead Times**

Place orders for products factoring in lead times from manufacturers and suppliers, including freight and customs clearance, as well as time for review and approval of shop drawings, samples, and tech data. Failure to order specified products on time will not be sufficient grounds to use alternative products. It is recommended that orders are placed as soon as practicable after contract award, subject to approval of submissions, and suppliers are provided with a delivery schedule which aligns with the contract completion date and construction schedule.

If specified products do not arrive in time for them to be installed prior to practical completion, then the contractor shall, at their expense, procure temporary fit-for-purpose products and then change them over to the specified products when they arrive, at a time convenient to the client. Commensurate amounts will be withheld by the superintendent until the installation of the specified products has been completed.

### **3.8 Uniformity**

All fittings, accessories and equipment of the same type shall be of the same manufacture and catalogue number.

### **3.9 Manufacturer's Warranty**

Provide manufacturer's warranty agreements where applicable.

Complete all necessary documentation on the Owner's behalf.

Include copies of warranties in the maintenance manuals.

### **3.10 Submissions**

#### **3.10.1 General**

Submit drawings, samples, technical data or other details as applicable for systems as specified, prior to ordering or installation.

Allow sufficient time for checking and return of submissions.

Co-ordinate submissions of related items.

#### **3.10.2 Drawings**

Provide hard copies as follows:

A1 size or larger – 3 sets

A3 size – 1set

Electronic - PDF

Drawings shall be submitted in original contract size, AutoCAD or REVIT format as applicable.

### **3.10.3 Samples**

Provide samples labelled to indicate type, part or catalogue number.

### **3.10.4 Technical Data**

Provide manufacturers technical specifications, type test report, performance data, installation details etc. as required.

### **3.10.5 Co-ordination**

Co-ordinate shop drawing design with other building and services elements.

## **3.11 Certification**

Certify that the works have been designed to the requirements of all applicable rules and regulations by providing a Form 15 to the satisfaction of the certifier.

Certify that the works have been installed to the requirements of all applicable rules and regulations to the satisfaction of the certifier.

Provide an electrical certificate of test at the completion of the project to certify that all electrical works undertaken as part of the project have been installed and tested in accordance with the respective state government acts, regulations and AS3000. Include a copy of the certificate of test in the O&M manual.

Provide a Form 16 at completion of the works. The Form 16 shall, at a minimum, reference all aspects certified in the designer's Form 15. Where particular aspects of the work to be certified are carried out by other subcontractors, each subcontractor shall provide a Form 16 as relevant to their scope of work.

## **3.12 Electromagnetic Compatibility**

Ensure that the entire installation covered by this specification meets the Electromagnetic Compatibility requirements of the Australian Communications and Media Authority.

## **3.13 Setting Out**

The positioning of equipment as shown on the Drawings is approximate and diagrammatic only, and these Drawings shall not be used for exact setting out. Locations of all equipment shall be confirmed on site by means of a formal and documented process such as drawing mark up, dimensioning or on-site mark up with as a minimum the Superintendent and Contractor in attendance. This co-ordination process shall be the responsibility of the Contractor to organise and it shall be carried out prior to rough-in.



### **3.14 Dedicated Electrical Circuits**

Liaise with main contractor and all other suppliers and subcontractors to confirm electrical requirements of all equipment and appliances prior to ordering or installing any cables.

Cable size, circuit breaker rating and type, and number of phases specified on electrical documents are based on equipment nominated by other disciplines. Suppliers and other subcontractors may select different equipment with different power supplies to those nominated in the design documents.

Adjust cable size, circuit breaker rating and type, and number of phases to suit the equipment being supplied. It is assumed that allowance for these adjustments is included in the main contract tender price.

### **3.15 Existing Inground Services**

Take all reasonable precautions to ensure that existing inground services are identified and protected from damage. Ascertain the locations of all other in-ground services prior to any excavation for services trenches or building work.

Engage an inground services locator to survey the entire works area including the building footprint and trenching routes prior to excavation. On the ground surface, mark the route and depth of all services identified and pothole to confirm where knowing their exact location and/or depth is critical. Engage a surveyor to plot the route and depth of all identified services to overlay on the as-constructed drawings.

Immediately notify the superintendent if any services are identified that may be problematic or which may have a time or cost implication.

### **3.16 Installation**

Install equipment and services square and in line with the building elements, plumb and at consistent mounting heights.

Where possible fix directly to the building structure.

Provide noggings, brackets, fixings and supports to manufacturer's recommendations and as required.

Where timber is used for supports or noggings, provide cypress pine or similar termite resistant species.

Organise reticulated services neatly and with applicable segregation.

Provide separate cable support systems for each reticulated service and support all cables over their entire route length.

Provide all access equipment and temporary lighting necessary for the installation of electrical services.

Allow for supports suitable for seismic (earthquake) restraint to AS 1170.4 where applicable.

### **3.17 Co-ordination**

Liaise and co-ordinate with all equipment suppliers to ensure that all equipment is installed and connected in accordance with the installation and functional requirements of the manufacturer. Provide ancillary equipment as required for correct system operation.

### **3.18 Seismic Restraint**

Provide equipment and equipment installation suitable for seismic restraint as required by NCC/BCA Sections BP1.1 and B1.2. The reference standard is AS1170.4 Earthquake actions in Australia, and the most relevant section is Section 8 Design of Parts and Components.

For this specification the seismic restraint pertains in particular to:

Support brackets

Fixings and fasteners, especially fixings into concrete and masonry

Cable trays and ladders and the attached cables, especially long runs where differential building movement could cause cable damage

Heavy equipment, especially where installed at high level relative to ground level, where installed overhead and where installed in ceiling spaces

Pipe runs subject to differential movement of building elements

Fixings for electrical switchboards

Equipment associated with safety systems and smoke control systems

Equipment and plant mounted on vibration isolating mounts

### **3.19 Environmental Conditions**

Ensure that all equipment is suitable for the environmental conditions to be encountered under normal circumstances for the site.

### **3.20 Building Penetrations**

Maintain integrity of building systems such as waterproofing and termite management at building penetrations.

### **3.21 Maintenance Access**

Locate all relevant equipment to ensure adequate maintenance access in accordance with Workplace Health and Safety regulations and building acts and code requirements.

### **3.22 Corrosion Protection**

All metallic materials and components shall be suitably protected against corrosion.

Bolts, screws, rivets, etc for external use shall be non-ferrous or stainless steel, or where approved galvanised steel.

Brackets, plinths, rods, etc shall be hot dip galvanised after fabrication.

All steel sheets shall be corrosion protected by means of degreasing, priming and painting as a minimum.

Where the application is in a highly corrosive environment such as in close proximity to the sea, provide suitable grade stainless steel or non-ferrous metallic components.

### **3.23 Finishes**

All equipment shall be painted unless the equipment material is chromium, anodised aluminium, plastics, stainless steel or other non-ferrous metals.

Paint and paint application shall comply with AS/NZS 2311 and AS/NZS 2312.  
Carry out all powder coating to AS3715.

### **3.24 Adhesives**

Provide adhesives and sealants capable of transmitting imposed loads, sufficient to ensure the rigidity of the assembly or the integrity of the joint and which will not cause discolouration or lack of adhesion of finished surfaces.

### **3.25 Fasteners**

Provide masonry anchors, bolts, nuts, washers, screws, nails and plugs as required for installation of electrical services.

Fastener materials, finishes and physical sizes shall be as appropriate for the loads, the types of materials in contact with the fastener and the environment.

Fasteners shall be selected to ensure full compliance with the requirements of Clause 3.20 Seismic Restraint.

### **3.26 Metals**

Provide metal sections and sheet as required for the electrical services.

Finish invisible joints by welding, brazing, soldering, grinding and buffing prior to application of finishes such as galvanising or painting.

Repair galvanised or electroplated metals using two pack organic primer to AS/NZS 3750.9.

### **3.27 Labelling**

Provide labels to equipment as required for identification, operation and warnings.

Labels shall be non-ferrous metal, multi-coloured laminated plastic or pre-printed flexible plastic.

Labels shall be of proven longevity without significant deterioration.

Externally mounted labels shall be weather and corrosion resistant.

Fix labels using nuts and bolts, rivets or self-adhesive material for flexible plastic labels.

Externally mounted fixings shall be corrosion resistant.

Warning labels shall be white lettering on red background with minimum 10 mm high lettering.

Minimum letter heights shall be 3mm.

### **3.28 Fire Integrity**

Where accessories, apparatus, cables or cable support systems pass through a fire rated building element, treat the penetration using tested and certified systems in accordance with AS 4072.1 and AS 1530.4 to ensure that the required Fire Resistance Level of the building element is maintained to comply with NCC BCA requirements.

Penetrations for cable support systems passing through horizontal or vertical fire rated building elements such as cable trays in ceilings and riser cupboards shall utilise fire-rated systems which allow the retrofit installation of additional cables without the need to modify or re-certify the penetration treatment. Products such as Trafalgar FyreBOX or similar meet this requirement.

Provide proprietary fire rated back boxes for switches, sockets and other such accessories mounted on fire-rated walls, installed in accordance with the manufacturer's instructions to retain the FRL of the building element in which they are installed.

Label and certify all penetrations through fire rated building elements and provide a penetration schedule in accordance with AS 1851 Section 17.

Refer to architectural drawings and any other relevant consultant documents to ascertain fire rated elements and the associated FRL.

### **3.29 Smoke Barriers**

Where cables or cable support systems penetrate smoke barriers, install and support this equipment to fully comply with all statutory requirements for smoke barrier building elements.

Label all smoke barrier penetrations and provide a penetration schedule in accordance with AS1851 Section 17.

### **3.30 Acoustic Integrity**

Install electrical services to comply with acoustic isolation requirements of the Building Code of Australia, AS/NZS 3000, Acoustic consultant report (where applicable) and any other statutory requirements.

Note in particular the requirements for residential units and buildings and provide acoustic rated wall boxes and light fitting covers as required to maintain the acoustic rating, which may need to be offset from other outlets so they are not within the same stud cavity.

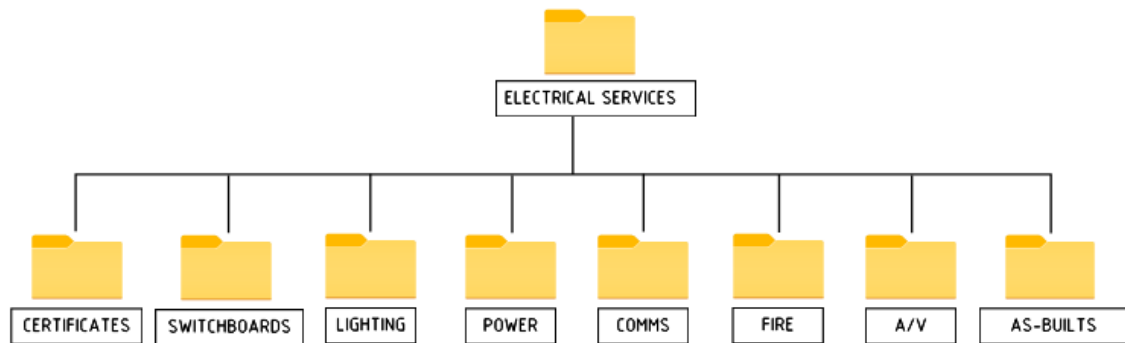
### **3.31 Operation and Maintenance Manuals**

#### **3.31.1 Manuals**

Provide Operation and Maintenance Manuals for all systems and components covered by this specification. Include As-Installed documentation for the entire installation in these manuals.

Once approved, provide one hard (paper) copy of manuals and drawings neatly bound into A4 size hard cover ring binders which are permanently labelled with the Project Name, Contract Description and Contractor's Name. Provide electronic copies on two USB storage devices.

Set up USB drive generally in accordance with the following file structure (with sub-folders for each scope item on the project), with relevant documents filed in the appropriate sub-folder:



Include the following in each set of manuals:

Contents page with page numbers for each section

Copies of all test results, certificates and approvals.

Manufacturer's operating instructions and details of model numbers, serial numbers, operating instructions, and manufacturer's guarantee certificate for equipment and appliances.

Service and user manuals for proprietary products and systems.

Emergency Lighting Log Book to AS 2293 requirements, as published by General Lighting Services Pty Ltd or equal approved.

Brief description of the Electrical installation and distribution network etc.

A recommended maintenance programme for all equipment installed.

A recommended list of spare parts.

As-Built Drawings (refer to As Built Drawing section)

Provide a draft set of the Operation and Maintenance Manuals to the Superintendent not less than 2 weeks before the date of practical completion, for the purpose of review and comment. The draft manual for approval shall be an electronic version submitted via electronic means (email or file transfer).

Submit final sets of the Operation and Maintenance Manuals to the Superintendent not more than 2 weeks after the receipt of comments on the draft from the Superintendent.

Final contract payment will not be approved until satisfactory Operation and Maintenance Manuals have been received.

### 3.31.2 As Built Drawings

Produce a full set of as-built drawings in REVIT and PDF format, which conform to the following general requirements:

Legend, site plan, schematic, and layout drawings in the same size and form as the original contract drawings but modified to reflect all changes which occurred during construction.

Remove or re-word notes that no longer apply to the “as built” state. For example, a note reading “Contractor to install garden lighting power supply in accessible ceiling space” should be re-worded to “garden lighting power supply installed in ceiling space” with an arrow showing the exact location.

Indicate locations of any equipment that may require inspection or maintenance, that has been installed in locations that are concealed from view (such as remote LED drivers and power supplies, lighting control modules, large marshalling/junction boxes etc)

Device numbering labels for all security and fire systems components, which match the field labels

Circuiting labels for all power circuits, lighting circuits and dedicated circuits

System schematic diagrams and schedules.

The title block of each drawing amended to include the Contractor's name and contact details, Contractor's drawing number and date and name of issue.

‘AS BUILT’ stamp applied to all drawings

### **3.32 Clean Up**

At completion clean all electrical equipment to as-new condition.

Remove all waste and debris on a continuous basis.

### **3.33 Completion**

All equipment and systems shall be fully operational and complete, all in accordance with manufacturers recommendations, prior to final testing and commissioning.

### **3.34 Testing and Commissioning**

Fully test and commission the complete electrical installation prior to practical completion.

Provide project specific Inspection, Testing and Commissioning Plans for all electrical systems. Only suitably qualified personnel shall carry out testing and commissioning. Provide evidence of qualifications when requested.

Record all test results and submit for inspection prior to practical completion.

### **3.35 Test Instrument Calibration**

All test instruments shall have current calibration certificates. Certificate copies shall be provided to the Superintendent prior to commencement of tests. Certificate currency shall be 12 months maximum.

### **3.36 Instructions and Training**

Provide skilled operators to instruct such persons as may be nominated by the Architect in the effective operation of all of the electrical systems in the installation. Instructions shall not be given until completion of all tests and adjustments.

Conduct training at agreed times and locations and for durations commensurate with the complexity of the systems.

## **4. SUPPLY AND METERING**

### **4.1 Definitions & Abbreviations**

|      |                                       |
|------|---------------------------------------|
| DNSP | Distribution Network Service Provider |
| NER  | National Electricity Rules            |

### **4.2 General**

Supply will be taken at low voltage from the point of supply:  
Padmount substation low voltage protective device

### **4.3 Supply**

Supply will be to Network Standards 3 phase, 230/400V, 50Hz, 4 wire, MEN earth system.

### **4.4 Distribution Network Services Provider (DNSP)**

The DNSP is:

Brisbane Airport Corporation

### **4.5 Standards & DNSP Requirements**

Comply with all regulations, standards and requirements of the Local Electricity Act & Regulations, NER, DNSP requirements, local council and BCA.

### **4.6 HV/LV Substation**

The DNSP is proposing to provide LV supply from a padmount transformer rated at 2000 kVA.

Liaise with the DNSP to ensure provision of all substation works and supply of electricity as required by the project program.

The DNSP will provide the following:

Padmount transformer

This contractor shall supply and install the following for use by the DNSP:

HV conduits from the site boundary to the transformer, including all trenching and backfilling, all to DNSP standards. Ensure all required inspections are carried out.

Pad mount transformer foundations, paving, thresholds, access, retaining walls, earthing arrangement and any other works as required by the DNSP to enable installation of their pad mount substation transformer.

Engage suitably qualified specialist contractors to perform the work as required by the local DNSP.

#### **4.7 Retail Metering**

Supply and install all equipment and enclosures as required for the provision of electricity metering. Carry out all works to DNSP Retailer requirements.

The general arrangement of metering facilities shall be as per Brisbane Airport Utility Metering Policy.

The project shall be installed with the EDM1 MK10 and EDM1 MK7 electrical meters connected to the BAC Ethernet Network to allow for connection to the Brisbane Airport Corporation's meter reading software (EDM1 – Multidrive FMR) and EDM1 Ezyview software for programming the EDM1 tariff meters. The meters shall be configured to a nominated tariff.

The programming of the meters shall be provided as a part of this contract and by the incumbent me

Refer to BAC TS-5900

The meter panel shall be of sufficient size to provide all appropriate minimum clearances to meet relevant standards. The meter panel shall be labelled as "BAC Metering".

CTs shall be sized to suit the maximum demand plus 25%.

All meters shall be connected to the BAC network via an RS485 connection.

Allow for sufficient meters for the project. All meters shall be provided with a calibration certificate.

Note: The connection devices from Ethernet to RS485 shall be Moxa RS485.

The EDM1 kWH meters shall be connected to water, recycled water, gas etc via control cables by this contractor.

#### **4.8 Applications**

Coordinate the actions required by all parties with Brisbane Airport Corporation to ensure the electricity service and all required metering is installed to meet project timeframes.



## **5. CABLES AND CABLE SUPPORTS**

### **5.1 General**

Supply and install all required cables and cable support systems.

### **5.2 Quality**

#### **5.2.1 Standards**

Installation and selection to AS/NZS 3000, AS/NZS 3008.1, AS/NZS 3003, AS/NZ 3013 and SAA HB301.

Cables to AS/NZS 5000.1, 2 and AS/NZS 4961.

Conduits to AS 2053 Series.

Conduits for telecommunications to AS/ACIF S008 and S009.

Fire and mechanical performance to AS/NZS 3013.

Testing to AS/NZS 3017.

### **5.3 Products**

#### **5.3.1 Power Cables**

Cables shall be multi-stranded copper conductor with PVC, XLPE and/or fire-rated insulation and sheath.

Cables shall be 0.6/1kV, V75 rated as a minimum standard.

Cables shall be colour coded as detailed in AS/NZS 3000.

Minimum cables sizes shall be:

Lighting sub-circuits: 2.5mm<sup>2</sup>

Power sub-circuits: 2.5mm<sup>2</sup>

Submains: 6mm<sup>2</sup>

Select cable sizes to comply with volt drop, short circuit and earth fault/touch voltage requirements of AS 3000.

Cable lugs shall be of the correct size and material for the conductor.

Cables for exterior lighting circuits shall be double insulated, external grade, suitable for use in underground conduit systems.

#### **5.3.2 Conduits**

Conduits shall generally be rigid non-metallic (UPVC or PVC), minimum 20mm diameter. Provide metallic conduits where shown or specified.

Conduits shall be to AS/NZS 2053 series.

Conduits for telecommunications systems shall comply with AS/ACIF S008 and S009.

Conduits for underground low voltage cabling shall be heavy duty rigid PVC to AS 2053.

Non-metallic conduits and fittings shall be grey or orange PVC as appropriate.

Metallic conduits and fittings shall be galvanised steel with screwed connections.

Provide flexible conduit only as required for connection of cables to equipment which is subject to movement or vibration.

Conduit fittings shall be of the same material and finish as the conduit.

### **5.3.3 Cable Ladders and Trays**

Cable ladders and trays shall be proprietary fabricated steel complete with all accessories and fittings for a complete tray system. Provide screw fixed lids as required.

Cable trays shall be perforated sheet steel folded and welded as required. Cable trays and accessories shall be zinc coated steel for internal use and hot dip galvanised for exterior use.

Cable ladders and accessories shall be welded steel construction, hot dip galvanised finish.

### **5.3.4 Skirting Wiring Duct**

Skirting wiring duct shall be extruded aluminium complete with clip in aluminium lid sections, internal segregation for multiple wiring systems and all necessary matching accessories for a complete system.

Finish shall be as shown on the legend or drawings.

### **5.3.5 Floor Outlet Boxes**

Floor outlet boxes shall be folded steel with cast metal lids, cable segregation and outlet mounting plates. Provide lid types and outlet quantities as specified. Plastic lids and surrounds will not be accepted.

### **5.3.6 Cable Pits**

Provide cable pits as required for installation of cables in the underground cable duct system.

Refer BAC guidelines for the required electrical and communication pits for use on the project

Pit lids shall be cast concrete, hot dip galvanised steel or trafficable cast iron and concrete filled.

Pits in trafficable areas shall be constructed to suit the expected loads and to Class C to AS 3996 as a minimum. Polymer pits shall not be provided for trafficable areas.

Pits in footpaths shall be fitted with cast iron, concrete filled lids, finished flush with the surrounding path.

### **5.3.7 Catenary Systems**

Catenary wires shall be multi-stranded galvanised steel type.

Fittings such as hooks, eyes and turnbuckles shall be galvanised steel.

## **5.4 Installation**

### **5.4.1 Cables and Cable Supports Generally**

Install low voltage cables in accordance with AS/NZS 3000 and in particular all of Section 3 “Selection and Installation of Wiring Systems”.

Install telecommunications cabling systems in accordance with AS/ACIF S009.

Maintain minimum installation clearances in accordance with AS/ACIF S009 and AS/NZS 3000.

Install all wiring systems (cables and their supports) concealed within the building fabric throughout unless in mechanical plant spaces, risers or as noted otherwise. Where it is proposed to run exposed wiring systems, obtain approval for the route and method before proceeding.

All wiring shall be fully supported along the entire route length.

Fix cable support systems to structural elements of the building only.

Cables run underground shall be installed in a conduit and pit system.

Cables and wiring for separate systems shall be run separately and on separate cable support systems or in segregated sections of the same support system.

Low voltage cable sizes shown on the drawings are based on installation methods to avoid derating in accordance with AS 3008.1 Tables 22 to 26. Increase cable sizes as required when cables are installed in a manner requiring a derating factor to be applied.

Run circuit earth cables over the same route and in the same wiring enclosure as the circuit active cables.

All cables shall be installed in a manner which enables easy removal and rewiring.

Cables in ceiling spaces shall be supported above the top of recess mounted light fittings.

All wiring shall be installed on the loop-in system with connections made at equipment or accessories.

Wiring in chases shall be run in concealed conduit.

Ensure minimum bending radius requirements for all cable types are maintained at all times.

For cables run in conduit use draw wires to pull in cable groups from point to point. Fill conduit to a maximum of 75% of the capacity noted in AS/NZS 3000.

Run all wiring systems grouped, tied or enclosed, square and parallel to the building structure.

Do not use adhesive fixing for cable support systems.

Run cables in external cavity walls in a manner which does not compromise the weather protective wall construction.

Support vertical cable runs to avoid damage due to cable mass.

Install cables in a manner that does not damage the cable during installation.

#### **5.4.2 Mains and Submains**

Run in conduits or on cable support systems such as cable ladder or tray.

Terminate using cable lugs and crimping tool.

Terminate submains to all switchboards provided by other trades (such as mechanical, hydraulics and lift services), unless noted otherwise.

Use suitable glands when entering equipment or assemblies.

Identify cables at both ends using stamped tags or similar.

Ensure that all cables are protected from accidental application of paint. Where cables have been painted the extent shall be assessed by the superintendent and cables shall be replaced at no cost as directed.

#### **5.4.3 Lift Submains**

Co-ordinate power supply and earthing requirements for lifts prior to installation. Install fire rated cables and low resistance insulated earth cables to ensure code compliance and lift supplier connection requirements.

#### **5.4.4 Fire and Mechanical Protection**

Where required, install cables and cable support systems with fire and mechanical protection (WS rating) to AS/NZS 3013.

Use stainless steel cable ties for all fire rated cables.

Provide fire rated wall boxes where required to maintain fire integrity.

#### **5.4.5 Conduits**

Conduits shall be rigid UPVC, PVC or metallic throughout except for short length final connections to direct connected equipment where smooth or corrugated flexible conduit may be used.

Install in straight, parallel runs, set at changes of direction, with maximum 90° bends, free of burrs and foreign materials. Provide draw-in boxes, expansion joints and saddles to ensure a conduit system free of visible sags and able to have cables pulled in after conduit installation. Draw in boxes shall not be located in inaccessible positions.

Conduit shall not be run against or in insulation.

Conduits run in reinforced concrete shall be run in the centre of the concrete slab or wall. Parallel runs of conduits shall have a minimum of 50 mm separation. Conduits cast into concrete shall be adequately secured to avoid movement or damage during pouring. Conduit ends shall be secured and capped to avoid damage and ingress of foreign material.

PVC conduits and fittings shall be joined using the recommended jointing adhesive.

Provide polypropylene draw cords or stranded and insulated draw cable (2.5 mm<sup>2</sup>) for all unused conduits.

For steel conduits paint ends and threads with zinc rich primer.

Use flexible conduit for connection to equipment subject to vibration.

All associated fittings shall be of the same material and colour as the conduit.

For telecommunications conduits:

Inspection fittings shall not be used

All bends shall be large radius type to ensure minimum bending radius conditions of cables are met

Apply temperature derating to AS/NZS 3080 where conduit is run in roof spaces.

#### **5.4.6 Cable Ducts, Trays, Ladders, Mesh and Skirting Duct Systems**

Install completed systems with all necessary lids, brackets, supports, fixings, rods and hangers.

Light duty cable tray shall only be used in concealed spaces or risers to carry and support final sub-circuits and ELV cabling systems (data, security, fire and MATV etc.).

Cable trays where run exposed shall be medium or heavy duty trays or cable ladders.

Groups of mains or submains cables shall not be run on light duty cable trays.

Cable tray lid shall not be used to support any cables.

For telecommunication systems, install to AS/ACIF S009.

Fixings, brackets and supports shall be fabricated from steel of suitable strength and rigidity to provide support to AS 1170.4.

Separate low voltage power cables by distances sufficient to avoid derating in accordance with AS/NZS 3008.1 Tables 22 to 26.

Comply with manufacturer's recommendations for support spacings based on maximum carrying capacity.

Size systems to ensure minimum cable bending radius requirements can be achieved at all times.

Provide clear space around cable support systems to enable installation of cables.

Provide segregation between cable support systems for different wiring systems. Ensure minimum separation of wiring systems as required by applicable standards and regulations.

Fix cables in suitable groups using cable clamps, ties or straps.

Provide bends with a minimum inside radius of 12 times the outside diameter of the largest cable installed.

Install clear of hot or wet equipment or surfaces in accordance with AS/NZS 3000.

Provide peaked galvanised steel lids over cable support systems where run exposed to sunlight and the elements.

Provide cable protection where cables enter or leave cable support systems.

Provide separate conduit access between skirting wiring duct compartments and ceiling space. Run conduits into proprietary cable access boxes located behind the skirting duct.

Ensure that 25% minimum spare capacity remains in all cable support systems at completion. Size all cable support systems accordingly.

#### **5.4.7 Catenary Systems**

Install using turnbuckles and intermediate ties to provide adequate cable support with minimal sag. Maximum 15m spacing between anchors and turnbuckles.

Fix cables using nylon cable ties, Velcro straps or similar approved system.

Run low voltage electrical cables on catenaries to avoid derating to AS 3008.1.

#### **5.4.8 Underground Systems**

##### **5.4.8.1 General**

Provide excavation, removal from site, rock excavation, conduits, pits, backfilling, reinstatement, cable markers and tracer wires for all underground cabling.

Provide separate underground conduit and pit systems for separate services (low voltage, communications, fire, etc).

Low voltage wiring systems shall be in accordance with AS/NZS 3000, Clause 3.11.

Cables shall not be installed direct buried.

Cables run in underground conduit systems shall be exterior grade. Power and lighting cables shall be insulated and sheathed. Single insulated cable shall not be accepted.

Record the route of underground systems prior to backfilling.

Set out conduit routes and obtain approval for these routes prior to excavation.

Provide polypropylene draw cords or insulated copper draw wires in all spare conduits.

Allow for final landscaped and finished levels when installing conduits and pits. Maintain minimum required cover. Pit lids shall be installed flush with the surrounding finished hardstand areas such as concrete or paving. Pit lids in landscaped areas shall be finished 50mm above the finished soil level to allow for mulching etc.

#### **5.4.8.2 Inspection and Hold Points**

Each trench section must be inspected for approval by the Superintendent prior to backfilling.

A photographic record may be acceptable for this purpose but it will be necessary to obtain Superintendent approval for this prior to proceeding past the hold points. Photographs of the trench sections shall show trench depth and calendar date.

Hold points shall be:

When the trench excavation is completed

When the services have been laid in the trench

Arrange for inspection by the Superintendent at each hold point.

#### **5.4.8.3 Excavation**

Saw cut or remove existing surfaces as appropriate.

Excavate trenches to the required depths and remove excess material from site.

Keep trenches free of water. Place bedding material, services and backfilling on firm ground.

Remove all sharp objects, rocks and organic material.

Give notice if rock is encountered to enable agreement on quantity.

Provide safety barriers to all open trenches.

Excavate and fill trenches in short sections to minimise safety hazards.

#### **5.4.8.4 Boring**

Engage a suitably qualified specialist contractor for all boring works. Fill all voids by pressure grouting.

#### **5.4.8.5 Conduits**

Provide bedding sand below, around and above conduits with minimum 50mm cover. Bedding sand under roads shall be stabilised with 3% cement mix.

Maintain separation between electrical, communications, Telstra, gas, water and other underground services to statutory requirements.

Provide conduit stubs for future extension where shown. Stubs shall be 1000mm and capped.

Use large radius bends for turned up conduits.

Use large radius bends and/or pits for changes of direction.

Install marker tape to AS/NZS 2648.1 above low voltage electric conduits.

Neatly cut conduits and remove all burrs and conduit filings so that cable stripping is avoided.

Secure all conduit sections using the recommended jointing adhesive.

Clean installed conduits free of debris and water.

Seal all conduits after cable installation using a product specifically manufactured for this application.

#### **5.4.8.6 Pits**

Provide sufficient pits to enable easy installation of cables.

The top of pits shall be finished flush with the surrounding surfaces.

Prior to installation, cut pit walls for conduit entries using separate circular holes for each conduit. Use suitable hole saws for this work. Pits cut in large sections for more than one conduit entry shall be rejected and these pits shall be replaced.

Seal around conduit entries after pit installation to prevent ingress of water and silt.

Drain pits to stormwater drainage system. Alternatively, obtain approval for rubble absorption trenches if draining to stormwater is not possible. Minimum rubble trench size shall be 300 x 300 x 2000 long using coarse gravel.

Install pits into a levelled and compacted excavation.

Found pits onto a wet concrete base at least 50mm thick or onto a compacted rubble base where the pit is drained to rubble.

The top of the pits and pit lids shall finish flush with surrounding surfaces in hardstand areas such as pavers, tiles and concrete. For landscaped areas allow for the pit lids to finish slightly above the finished soil areas to allow for turf, mulch or similar landscaped finish.

Cut conduit entries not more than 50mm past the inside faces of the pit walls.

Provide rigid conduit bracing bars inside the pits to prevent wall collapse.

Provide poured concrete haunches (minimum 150mm thick) around multiple conduit entries.

For pits in trafficable areas (Class C and above) backfill around the pits to the finished surface with concrete (minimum thickness 100mm).

For all other pits backfill around pit using compacted bedding sand and finish at the surface perimeter of the pits with a 200 wide and 200 thick in-situ poured and trowelled concrete collar, reinforced as necessary.



#### **5.4.8.7 Backfilling**

Backfill trenches and compact using a ramming machine in maximum 200mm layers. Complete the backfill with a 100mm minimum layer of topsoil.

Backfill material shall not contain any rocks, organic material, or sharp objects.

Backfill under roads and paved areas shall be fine crushed rock compacted in 150mm layers to 100% STD compaction.

#### **5.4.8.8 Reinstatement**

Existing surfaces shall be reinstated to their original condition.

Lawn areas – Reinstatement removed lawn segments or provide new turf to match existing.

Paving – Lay pavers on sub-base to match existing. Provide additional pavers as required.

Concrete – Key new poured concrete into adjacent existing concrete using reinforcing steel grouted into drilled holes.

Roads – Provide rock sub-base, base and bitumen to match existing.

#### **5.4.8.9 Route Markers**

Provide route markers to accurately indicate the location of underground conduit routes.

Install markers on pits, at 50 m intervals on straight runs, at changes of direction and at ends of run.

Markers shall consist of 16mm thick engraved stainless steel arrows, discs or rectangles as appropriate showing direction and type of service. Fix markers to concrete base using stainless steel friction pins with mushroom head.

#### **5.4.8.10 Trench Warning Tape & Trace Wire**

Install 100mm wide detectable trench tape in all electrical and communications trenches above each group of corresponding service conduits. The tape shall be installed at approximately 200-300mm below finished ground level.

Trench tape to contain words to the effect of:

“WARNING ELECTRIC/COMMUNICATIONS CABLE BELOW”

Trench tape to have 316 stainless steel wire inlaid, in a wave pattern to facilitate stretching and to improve detectability, laminated between two layers of low density, lead free rot-resistant polyethylene tape.

Trace wires shall be joined using approved crimp connectors to ensure continuity.

Trace wires to be terminated at the start and end of each trench route and at each pit using appropriate clamps in accessible locations, fixed to pit walls or building elements. Trace wires should be earthed to manufacturer’s recommendations.

Trench tape to be Tapex Wavelay or similar, installed in accordance with manufacturer's instructions.

#### **5.4.8.11 Cables**

Install cables within conduit systems in accordance with manufacturer's instructions.

The cable cross sectional area shall not exceed 40% of the internal cross sectional area of each conduit. Provide additional conduits as required.

Run individual submains circuits in separate conduits.

Cables with insulation or sheathing damaged during installation shall be removed and replaced.

Run insulated and sheathed cables for external light and power circuits.

#### **5.4.8.12 Surveyed As-Built Drawings**

For underground conduit systems provide surveyed as-built drawings showing exact routes and locations, and include in the as-installed drawing set in the maintenance manuals.

### **5.5 Completion**

Carry out all tests as detailed in AS/NZS 3000 Section 8 "Verification".

Use AS/NZS 3017 "Electrical Installations – Testing and Inspection Guidelines" as the basis for test procedures.

Record all test results and include in Maintenance manuals.

## **6. EARTHING**

### **6.1 General**

Supply and install Protective and Functional earth systems to ensure safe operation of the electrical installation.

The protective earth system shall be multiple earthed neutral (MEN), or TN system to IEC60364.

Liaise with the DNSP to determine the earthing system at the substation, combined HV/LV or separated.

Provide substation earthing facilities to DNSP standards.

### **6.2 Quality**

#### **6.2.1 Standards**

AS/NZS 3000 Section 5 “Earthing Arrangements and Earthing Conductors” for electrical system earthing.

AS/NZS 3003 for patient areas of hospitals, medical and dental practices.

AS/NZS 2381 series and AS/NZS 61241 series for hazardous locations.

AS/NZS 1768 for lightning protection bonding earths.

AS/ACIF S009 for telecommunications cabling functional earths and equipotential bonding.

### **6.3 Products**

Provide earthing system components manufactured from copper, brass, stainless steel or copper clad steel for earthing electrodes.

Provide equipment as required for the protective earth system, functional earths, telecommunications system connection to earthing system and equipotential bonding.

Provide guards, covers, pits and enclosures as required to ensure mechanical protection of the complete earthing system.

Earth rods shall be copper clad steel suitable for direct complying. Provide driving tips and couplings as required.

Provide labels for earthing components as required.

### **6.4 Installation**

#### **6.4.1 General**

Protect the complete earthing system from mechanical and environmental damage.

Ensure all earthing system components are sized to resist the electrical forces (fault currents). Provide calculations where required. Refer to AS/NZS 3000 Clause 5.3.3.1.2.

Earth all exposed conductive parts of the electrical system.

Earth all conductive building materials and structural metalwork, including reinforced concrete for pools, spas, and floors of bathrooms, safety showers and similar wet areas.

Provide separate earth cables for each circuit.

Install earth electrodes in locations which ensure effective contact with moist soil and are separated from conductive enclosures of other buried services such as water, gas and telecommunications.

Where high soil resistivity is encountered provide drilled cores with bentonite resistivity enhancer and multiple earth electrodes as required.

Bond incoming conductive enclosures of buried services such as water, gas and telecommunications to the protective earthing system. Refer to AS/NZS 3000 Clause 5.6.

Provide equipotential bonding as required for other systems or equipment as noted in AS/NZS 3000 Clause 5.6 and related standards for the particular systems.

Ensure continuity of earthing connections to all earthed systems and structures.

Conduct visual inspections and electrical earth continuity testing of equipotential bonding of steel reinforcing prior to concrete being poured.

Provide telecommunications system earthing to AS/ACIF S009, related ACMA standards and AS 3080.

#### 6.4.2 Main Electrical Earth

Provide a low resistance main earth bond to the general mass of earth appropriate for the site (near the MSB and / or near each MEN point).

Provide accessible earth rod inspection pits for earth electrodes and label in accordance with AS/NZS 3000. Earth rod inspection pits shall be ERITECH PIT-03 or approved equal.

Determine soil resistivity conditions either by measurement or visual inspection of the site. Resistivity of typical soil types are as follows:

|                |                   |
|----------------|-------------------|
| Clay           | 8 to 70Ωm         |
| Loam           | 5 to 250Ωm        |
| Sand           | 200 to 3,000Ωm    |
| Gravel or Rock | 3,000 to 30,000Ωm |

For sand, gravel / rock or very dry soil conditions provide a project specific main earth solution.

Ensure that the earth grid electrodes are installed into the original natural soil. Installation into any type of fill material shall not be permitted.

Ensure that the earth grid is installed into soil which is naturally moistened as a result of weather activity. (Note that the resistivity of a sample sand / clay mix can vary from 1,500Ωm at 2.5% moisture content to 40Ωm at 30% moisture content).

Provide as a minimum in favourable soil conditions such as loam or clay the following main earth electrode arrangements:

|                        |                                                                                                                                                                                                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <63A supply            | One 1440mm long driven copper clad steel earth rod.                                                                                                                                                                                                                                  |
| >63A up to 400A supply | Two 1440mm long driven copper clad steel earth rods, either coupled for a total driven depth of 2880 or separately driven with a minimum separation of 2m and bonded with a 95mm <sup>2</sup> copper, insulated earth cable and proprietary earth rod clamps finished with cold gal. |
| >400A supply           | A minimum of two 2880mm long (2 x 1440mm each) driven copper clad steel earth rods spaced a minimum of 3m apart and bonded with a 95mm <sup>2</sup> copper, insulated earth cable and proprietary earth rod clamps finished with cold gal.                                           |

For less favourable soil conditions such as sand, gravel / soil loose mixtures and solid rock, provide soil resistivity enhancing material around each earth rod (Erico Gem or equal). Provide earth rods of increased quantity, spacing and depth as required to achieve a minimum measured resistance to earth of 10Ω.

## 6.5 Specific Earthing Arrangements

Provide large cross-section earth conductors as required for specific installations such as lifts and UPS. Confirm required earth cable size with equipment manufacturer.

## 6.6 Completion

Inspect and test the earthing system in accordance with AS/NZS 3000 Section 8 “Verification” including all applicable mandatory tests (Clause 8.3.3), continuity of the earthing system (Clause 8.3.5) and earth fault loop impedance (Clause 8.3.9).

## **7. SWITCHBOARDS**

### **7.1 General**

Supply and install all switchboards as required with the minimum shown on the drawings.

Switchboards shall be constructed by a specialist switchboard manufacturing company. Refer to the approved list of manufacturers in Section 2 of this Brief.

Switchboards shall be fully enclosed, metal cubicle, circuit breaker type unless shown otherwise.

### **7.2 Quality**

#### **7.2.1 Standards**

BAC Standards and Guidelines

AS/NZS 61439 Series for switchboard design, verification, and construction

AS/NZS 60898.1 and AS 3111 for miniature circuit breakers.

AS 60947 Series for low voltage switchgear and control gear.

AS/NZS 3190 for RCD circuit breakers.

AS/NZS 3760 for In-service safety inspection and testing of electrical equipment.

AS 61818 series for HRC fuses.

AS 60044.1 for current transformers.

AS 4070 Category 11 for surge diverters (transient protection).

DNSP for metering provisions and general arrangement.

#### **7.2.2 Submissions**

##### **7.2.2.1 Type Test/Verification Data**

Submit test & verification documentation in the name of the switchboard manufacturer for assemblies and components to demonstrate compliance.

##### **7.2.2.2 Proprietary Modular Switchboard System Accreditation**

The switchboard assembler must have current accreditation & certification from the modular switchboard system Original Equipment Manufacturer (OEM), stating they are trained and accredited to design and construct the nominated modular system of assemblies that are proposed.

The switchboard assembler must have documented quality assurance procedures in place to verify that the modular assembly is designed in complete accordance with the OEM's instructions, and then verified that the assembly has been constructed in complete accordance with that verified design.

Copies of these documents shall be produced on request.

##### **7.2.2.3 Shop Drawings**

- Submit shop drawings showing:
- Types, model numbers and ratings of assemblies.

- Component details, for all functional units and transient protection.
- Detailed dimensions.
- Verification details for particular assembly components.
- Shipping sections, general arrangement, plan view, front elevations and cross-section of each compartment.
- Projections from the assembly that may affect clearances or inadvertent operation, such as handles, knobs, arcing-fault venting flaps and withdrawable components.
- Fault level and rated short circuit capacity characteristics.
- Ingress Protection (IP) rating for the Assembly and for the internal barriers.
- Fixing details for floor or wall mounting.
- Pollution degree
- Front and back equipment connections and top and bottom cable entries.
- Door swings.
- External and internal paint colours and paint systems.
- Quantity, brand name, type and rating of control and protection equipment.
- Construction and plinth details, ventilation openings, internal arcing-fault venting and gland plate details.
- Terminal block layouts and control circuit identification.
- Single line power and circuit diagrams.
- Details of mains and submain routes within assemblies.
- Busbar arrangements, links and supports, spacing between busbar phases and spacing between assemblies, the enclosure and other equipment and clearances to earthed metals.
- Dimensions of busbars and interconnecting cables in sufficient detail for calculations to be performed.
- Form of separation and details of shrouding of terminals
- Labels and engraving schedules.

#### 7.2.2.4 Calculations

Submit design calculations to verify design assumptions, in accordance with AS/NZS 61439 Series.

Submit design calculations to verify busbar ratings.

Submit protection device grading calculations for the electrical distribution system to demonstrate full discrimination and AS3000 compliance for all sources of electricity supply (ie Authority grid and/or generator).

#### 7.2.3 Design Criteria

Construct switchboards to comply with the following:

Rated busbar current:

As shown on drawings for Main Switchboards

160A (minimum) or as shown on drawings for distribution boards

Full size neutrals

Rated prospective short circuit current:

Greater than calculated fault level at switchboard location

As shown on drawings for Main Switchboards and as calculated by manufacturer

6 kA (minimum), or as shown on drawings for distribution boards

Degree of Protection:

IP42 to AS 60529 for internal switchboards in a dedicated switchboard cupboard or switchboard room

IP56 to AS60529 for switchboards located in a path of travel for containment of smoke (NOTE: applies to any switchboard that is not in a dedicated switchboard room or cupboard)

IP56 to AS 60529 for external switchboards

IP2X between adjacent compartments in switchboards (after cable installation)

Arc Fault Containment

Form of Separation:

- Form 4a for Safety Services
- Form 3a for Main Switchboards
- Form 3b for all switchboards rated at 800 Amps or more (in accordance with AS/NZS3000 Section 2.5.5)
- Form 1 for Distribution Boards

Physical Size:

Designed for installation into the spaces allocated on the drawings

Future Circuit Spare Capacity:

25% or as shown on drawings, whichever is the greater

Grading of Protection Devices:

Designed to ensure full co-ordination with calculated grading from DNSP protection through to sub-circuit protection i.e. full discrimination

Design protection grading for both Mains and Generator supplies where applicable.

Busbar Temperature Rise Limits:

Maximum rated current – temperature rise 50°C, ambient temperature 40°C

Maximum short-circuit current – temperature rise 160°C

Limitation of Internal Arcing Fault Currents:

Comply with AS/NZS3000 Section 2.5.5.

Uniformity

Provide circuit breakers from one manufacturer to ensure complete protection grading.

Pollution Degree

3 for Main Switchboards rated at more than 250A

2 for distribution boards

RCD Rapid Test facility

All switchboards shall have a Rapidtest facility installed in all switchboards to test the installed RCD circuit breakers

Zencontrol Space



All internal and building mounted lights shall be controlled by Zencontrol and the switchboards shall have sufficient space within them, in a separate compartment for DIN rail mounted Zencontrol modules

#### **7.2.4 DNSP Requirements**

Incorporate all DNSP equipment and requirements into the switchboard assemblies.

Obtain DNSP approval as required for the design prior to fabrication.

#### **7.2.5 Verification and Testing**

Provide verification details for Assembly construction (PTTA to AS3439.1). Factory test and verify Assembly for Routine test in accordance with AS/NZS 61439.2.

Provide copies of these tests and verification documents on completion.

### **7.3 Products**

#### **7.3.1 General**

Switchboards shall be fully enclosed, metal cabinet type, proprietary construction for distribution boards (typically) and custom built for main switchboards.

Front access assemblies with frontal area less than 2m<sup>2</sup> shall be wall mounted. All other assemblies shall be floor mounted.

Design assembly cabinets to enable transportation to site and positioning into the available space during construction.

#### **7.3.2 Assembly Cabinets**

Provide rigid, square enclosure cabinets of fabricated sheet metal, folded and welded construction and of sufficient structural rigidity to ensure transport to site and installation without distortion or damage.

Design the assembly cabinet to fit in the space provided, ensuring compliance with AS3000 2.10 for clearances around the assembly.

Indoor assemblies shall be 1.6mm (minimum) zinc or galvanised mild steel, powder coated finish.

Outdoor assemblies shall be either 2.0mm (minimum) 316 stainless steel brushed finished, or 3.0mm (minimum) marine grade aluminium, powder coated finish. All outdoor assemblies shall be fully seam-welded for exterior joins – assemblies relying on bolted joins with gaskets or sealant to achieve an exterior IP rating will not be accepted.

Machine fold sheet metal angles. Weld all joints and finish flush with adjacent surfaces. All bare edges shall be lipped with minimum return of 13mm. Provide stiffening members as required for rigid construction.

Support frames and structural members shall be cold rolled or extruded steel sections welded or bolted to the cabinet metalwork. Steel for outdoor assemblies shall be hot dip galvanised.

Provide degree of separation as specified by means of horizontal and vertical steel partitions, and insulating barriers, insulation and housings where a “bih” or “ih” form rating has been specified. Heat shrink fixed with cable ties is not acceptable for this purpose.

In addition, provide panels in vertical compartment walls for controlled cable and busbar access between adjacent compartments. Busbars and cables shall penetrate the vertical panels at right angles. Provide adequate seals and cable glands to maintain the Degree of Protection and Form of Separation.

Ensure vermin proof construction throughout. Cover ventilation openings with non-ferrous 1mm mesh.

Fix equipment or components to panels or support frames using corrosion resistant bolts, studs, set screws or proprietary clips. Fixings shall be accessible after completion to enable future equipment changes.

Lifting provisions shall be included for assemblies larger than 1800mm high x 600mm wide, or as required for transportation and installation.

Provide mounting structure and fixings which allow assemblies to be fixed in position without removing equipment.

Floor mounted assemblies shall be provided with hot dip galvanised steel plinth, minimum 75mm high, 40mm wide, 6mm thick, drilled and bolted to assembly base and floor or plinth.

### **7.3.3 Doors, Removable Panels and Escutcheons**

Provide lockable doors, removable panels and escutcheons for complete assemblies.

Fabricate from the same material and construct as for the assembly cabinet.

Maximum panel widths shall be 800mm. Minimum door and escutcheon swing shall be 90°. Maximum cover and escutcheon area 1.2m<sup>2</sup>.

Doors and covers shall have right angle returns on all sides fitted with continuous sealing rubber gasket fixed on a metal retaining frame with industrial grade adhesive.

Provide stays to exterior mounted assembly doors.

Ensure door seal contact onto the assembly flat surface and for exterior mounted assemblies the door seal shall be outside a gutter formed around the full perimeter of the assembly opening. “French door” style arrangements are not acceptable.

Provide chrome plated non-ferrous door furniture of sufficient strength to withstand fault condition explosive forces within the assembly.

Provide doors, covers and escutcheons as follows:

#### Doors

Lift off 316 stainless steel hinges with unequal pin lengths

Min. of 3 hinges for doors higher than 1000mm  
Selectrix swing model 1107-SS02-316 for all external handles.  
Keyed locks to BAC requirements.

#### Covers

Lift off hinges as for doors  
D handles for removal and replacement

#### Escutcheons

Hinged  
Knurled thumb screws with locating pins or quarter turn slotted cam locks, minimum 4 fixings  
D handles for removal and replacement

Maintain earth continuity to all doors, covers and escutcheons.

### **7.3.4 Cable Zones**

Provide vertical cable zones between each compartment tier containing electrical switchgear on busbars or chassis' rated greater than 500 amps. Compartments containing double-sided chassis shall have cable zones down both sides of the compartment.

Where bottom entry plinth-mounted switchboards do not have a cable pit directly below them, provide cable zones at the base of the enclosure to facilitate access to the conduits rising beneath the switchboard to enable future hauling of cables through these conduits.

Cable zones shall be a minimum of 300mm wide, or the width required to permit the bending radius of outgoing cables used, whichever is the greater.

Cable zones shall include suitable cable supports to ensure there is no stress or strain on cable terminations, switchgear, or earth/neutral bars.

Cable zones shall be provided with dedicated hinged access doors.

There shall be no exposed live parts in a cable zone (any live parts able to be touched with a standard test fingers are deemed to be exposed).

### **7.3.5 Cable Entries**

Provide cable entries for incoming and outgoing cables.

Provide sufficient clear space to allow installation of cables, including cables for all future circuit provision, without undue bunching or sharp bends.

Provide removable non-ferrous gland plates as installed to maintain the degree of protection.

At cable entries run all cables through neat-fitting compression glands. All cable entries shall be sealed against the spread of fire and smoke, and sealed against the ingress of moisture, debris and vermin.

Provide cable support systems within the cables zones for fixing of cables. Size cable supports to allow for future circuits to the capacity of the assembly.

### 7.3.6 Finishes

Apply protective coatings to internal and external metal surfaces of assembly cabinets (except for external surfaces for stainless steel). Internal surfaces includes escutcheons, and all interior surfaces of the switchboard enclosure.

Paint finish shall be thermoset powder coating to AS3715 and APAS-0155/1 with finish colours as follows:

Indoor Custom/Modular Assemblies:

External surfaces – Orange or Grey

Internal surfaces – White

Outdoor Custom Assemblies:

External surfaces – Grey (brushed for stainless steel)

Internal surfaces – White

Final Sub-circuit Distribution Boards:

External surfaces – Orange or Grey

Internal surfaces – White

Note: If other colours are noted on the drawings, the colours on the drawings take precedence.

### 7.3.7 Busbars

Provide solid busbars within assemblies from the incoming supply terminals to the line side of protective equipment for outgoing functional units, including future functional units.

For switchboards installed against a wall, all busbars shall be installed so that they are accessible and inspectable along their entire length, from the front of the switchboard. This is to facilitate future additions, inspections, thermal scanning, and maintenance. Busbars installed at the rear of switchboards behind metal partitions within the assembly will be rejected. Rear-access busbars will only be permitted where the switchboard is installed with a minimum of 1 metre clear access behind the switchboard, and removable panels are fitted to all compartments containing busbars.

Busbars shall be solid electrical grade hard drawn copper, rectangular section, tinned.

Use of flex busbar and cables in lieu of solid busbar is not acceptable unless specifically noted on the drawings.

Neutral busbars shall be of the same current carrying capacity as the active busbars.

Busbar ratings shall be as noted on the drawings or as detailed in the design requirements.

Tee-off busbars shall be rated to the maximum frame size rating of the functional unit or the diversified rating for multiple functional units on the one tee-off.

With the exception of busbars that are within a dedicated busbar zone, all busbars shall be fully insulated by colour coded insulation using non-deteriorating insulation material.

Busbars in main switchboards shall be vertical laid to facilitate cooling. Horizontal lay for busbars will be rejected except for short runs such as tee-offs. For busbars rated at 1000A and above spacing between phases shall be a minimum of 100mm.

Support busbars on non-hygroscopic insulating material designed to withstand temperature rise and fault current parameters.

Busbars passing through slots shall be sleeved with plastic shrink type sleeving.

Provide fully insulated and colour coded busbars for distribution boards.

Provide colour coded bands at least 50mm wide on uninsulated busbars in main switchboards. Locate on each busbar section in every compartment. Do not use adhesive tape.

Colour code busbars as follows:

Active: red, white, blue

Neutral: black

Earth and MEN link: green/yellow

Maintain phase sequence throughout the assembly.

Run neutral and earth busbars into each compartment containing functional units. Number all terminations. Provide terminals for all future circuits. For cables greater than 10mm<sup>2</sup> provide bolts or studs. Tunnel type terminations are acceptable for cables up to 10mm<sup>2</sup>.

Where functional units of the switchboard are rated at 1000Amps or greater, the neutral busbars shall follow the active busbar through the functional units via the same gland plates.

Join busbars using high tensile bolts and nuts, locked in position and tightened to the recommended torque using a tension wrench. Busbar joints shall be fully lapped to maintain adequate joint surface area.

Busbars marked by scribing will be rejected. All marking shall be pencil or ink.

Provide pre-drilled main circuit supply busbar for future extension where shown.

Extend busbar tee-offs to all future functional unit locations.

Provide removable busbar sections less than 400mm long for current transformers.

For MEN links greater than 10mm<sup>2</sup> provide bolted removable busbar link. .

Provide busbar flags for equipment with terminals too small for cable lugs. Support busbar flags as required. Provide barriers where minimum clearance distances are not satisfied.

Pre-drill the main busbar for the future extension and extend busbar droppers to spare locations. Drill each dropper to suit connection of future equipment of the same type as that specified.

Distribution board busbar chassis pole quantities are based on 1 pole position for integral MCB/RCD's. Increase pole quantities if integral MCB's/RCD's take up more than 1 pole location.

### **7.3.8 Wiring**

Provide all required internal wiring in switchboard assemblies with insulation grade appropriate to the design temperature at terminations.

Cables shall be sized for current rating based on the internal ambient temperature and also fault current withstand capability.

Run cables neatly and support clear of busbars and functional units.

Identify cables at both ends.

Shroud terminals where exposed, particularly on hinged panels and doors.

Identify all internal instrumentation and control cable ends by clip on markers. Adhesive markers shall not be acceptable. Indicate numbering on "As Installed" circuit diagrams. No numbers shall be duplicated.

### **7.3.9 Facilitation of Safe Inspection & Maintenance**

Switchboards must be designed, manufactured, and installed in a fashion that facilitates safe in-service inspection and maintenance. This includes competent persons conducting inspections, testing, and thermographic scanning.

Irrespective of, and in addition to the specified Form Rating of the switchboard, all live conductors in all compartments that contain functional units shall be provided with insulating barriers or housings arranged in such a fashion that a person opening the escutcheon is separated from live parts to IP2x (standard test finger).

This may be achieved by the following means:

Insulated Busbars & chassis

Clear acrylic barriers

Proprietary termination shields and phase barriers for switchgear

Barriers and shield shall be secured in place and shall be able to be easily removed and replaced for testing purposes. Cables ties are not an acceptable means of securing in place. Screws or nuts & bolts are acceptable.

Spliced, wrapped, and cable tied heat shrink is not acceptable and will be rejected.

Switchboard shall be manufactured and supplied with all means for compliance with this requirement, along with any instructions required for the installing electricians to follow.

### **7.3.10 Equipment Layout**

Arrange equipment in logical positions to provide a neat and simple design.

Locate equipment to minimise derating due to temperature rise and to ensure that the design temperature is not exceeded at each functional unit and inside the assembly.

### **7.3.11 Exterior Mounted Switchboards**

Additional requirements for externally mounted switchboards include:

Weather and sun shading tilting “roof” located 50mm above the assembly top and extending out 100mm on front and sides.

Bottom cable entry only.

All hinges, door hardware, handles, bolts etc shall be 316 stainless steel.

3-point locking cam/bar/roller latching system for all doors

Hooded ventilation openings top & bottom, with vermin proof stainless steel mesh

Custom made concrete cable pit and concrete slab for mounting switchboard, with steel access covers to access cable pit below.

### **7.3.12 Circuit Breakers**

Provide circuit breakers of Schneider or NHP Terasaki, manufacture.

For loads to 63A provide 18mm wide miniature circuit breakers (MCB’s) (where fault level & outgoing cable size permits).

For loads 63A to 125A provide 27mm wide miniature circuit breakers (MCB’s) (where fault level & outgoing cable size permits).

For loads greater than 125A up to 1600A, or where cable size is too large for miniature circuit breaker terminals, or where fault level exceeds the rating of a miniature circuit breaker, provide moulded case circuit breakers (MCCB’s).

For loads greater than 1600A provide air circuit breakers.

#### **7.3.12.1 Miniature Circuit Breakers (MCBs)**

Provide miniature circuit breakers of the specified fault and current rating complying with AS/NZS 60898.1 and AS 3111.

Miniature circuit breakers shall be of the same manufacture throughout.

Mount miniature circuit breakers to manufacturer’s requirements and to ensure that 1, 2 or 3 phase MCBs are interchangeable onto the same busbar chassis.

Align operating toggles in the same plane.

#### **7.3.12.2 Residual Current Operated Circuit Breakers (RCD/MDBs)**

Provide combined RCD/MCBs of the same manufacture as miniature circuit breakers, complying with AS/NZS 3190 and with a maximum RCD tripping current of 30mA.

NOTE. NHP Rapid Test facility shall be installed in all switchboards that contain RCD’s

Provide RCD/MCB’s which use the same poles and pole spacing as MCBs.

Select the correct type of RCD for the circuit in accordance with AS3000 Clause 2.6.2.2, manufacturer's advice for specific equipment, and the following table provided for general guidance:

| RCD Type:            | AC                 | A                   | B                | F                |
|----------------------|--------------------|---------------------|------------------|------------------|
| Loads:               | AC-only loads      | Comb. AC & DC loads | Electronic loads | Electronic loads |
| Typical Use Examples | Hot water systems  | Lighting circuits   | Solar inverters  | Solar inverters  |
|                      | Heaters            | SSO Circuits        | Data centre      | Inverter air con |
|                      | AC fan motors      |                     | EV Chargers      | Data rack/UPS    |
|                      | Cooking appliances |                     | Inverter air con | EV Charger       |

#### 7.3.12.3 Moulded Case Circuit Breakers (MCCBs)

Provide MCCBs of the specified fault and current rating complying with AS 60947.1 and AS 60947.2.

Derate MCCBs as required for installation conditions.

Use MCCBs of the same manufacture throughout.

Align operating toggles in the same plane.

Mount MCCBs in accordance with manufacturer's recommendations.

Trip units for MCCBs rated at 160A or greater shall be adjustable electronic type.

#### 7.3.12.4 Air Circuit Breakers

Provide air circuit breakers to AS 60947.1 and AS 60947.2, open construction, withdrawable with integral solid state protection relay.

#### 7.3.13 Switches/Isolators and Fuse Switches

Provide units complying with AS 60947.1 and AS/NZS 60947.3 and rated to the enclosure installation conditions and anticipated loads.

Provide locking facilities in the OFF position.

Shall be from the same manufacturer as for all other switchgear.

#### 7.3.14 Contactors

Provide contactors to AS 60947.4.1 rated for the enclosure conditions and anticipated loads.

Shall be from the same manufacturer as for all other switchgear.

#### 7.3.15 Fuses with Enclosed Fuse Links

Provide HRC type fuses and fuse carriers to AS 61818 series.



### **7.3.16 Current Transformers**

Provide current transformers complying with AS 60044.1, cast resin encapsulated window type with busbar clamps.

Provide test links and secondary shorting links in accessible positions.

Accuracy Class shall be:

Energy measurements: Class 0.5

Indicating instruments: Class 3

### **7.3.17 Transient Protection**

#### **7.3.17.1 Surge Diverters – Main Switchboard**

Provide distribution switchboard parallel surge protection on Main Switchboard as follows:

240v nominal system voltage

275v max continuous operating voltage

MOV and/or gas discharge type with visual indication, replaceable modules and alarm contacts

Maximum Discharge Current, I<sub>Max</sub>, 100kA per line 8/20us

Nominal Discharge Current, I<sub>n</sub>, 40kA per line 8/20us

Three mode protection; L-E, L-N, and N-PE

Mechanical flag status indication

Overcurrent fuses as per manufacturer recommendations

Must be selected from either Crittec/Erco, Schneider, Novaris, DEHN, Eaton, or NHP Cirprotec manufacture.

### **7.3.18 Energy Check Metering**

Provide energy metering as shown on schematics to comply with BCA Part J8.3.

Provide meters, current transformers, voltage fuses and all other ancillary devices required.

Meters shall be single input, or dual input for split load groups, such as measuring lighting and power energy separately with a single meter. Dual input meters must have separate data capture for each load group, and the ability to scroll through the display to view each load group's information.

Meters shall measure for each load group:

Voltage (L-L and L-N)

Current (amps)

kW

KVAr

KVA

Maximum Demand

kWh

kVAh

Meters shall incorporate a Modbus RS485 data output (and connect to energy monitoring system where specified).

Meters shall be Carlo Gavazzi EM270 or similar equivalent.

### **7.3.19 Control Devices and Switches**

#### **7.3.19.1 Rotary switches**

Provide cam operated type with switch positions arranged with displacement of 60°. Locate off position at 12 o'clock. Provide rectangular escutcheon plates securely fixed to the assembly panel. Identify switch position and function.

#### **7.3.19.2 Time switches**

Provide simple DIN mount mechanical analogue time switches with:  
Long life battery backup  
Minimum contact rating of AC1 16A  
15 minute time block adjustments

Time switches shall be NHP Finder MECHD1211B, Schneider ACTI9 ARM, Hager EH111 or approved equal.

Time switches shall be mounted so that they can be adjusted by any person without opening switchboard escutcheon panels.

#### **7.3.19.3 Control relays**

Provide control relays as follows:  
Suitable for continuous operation. Provide relays selected in conformance with the load duty.  
Plug-in type with bases with captive clips which can be operated without using tools.  
Electrically separate, double break with silver alloy, non-welding contacts.  
With contact blocks readily convertible to either normally-open or normally-closed contacts.

#### **7.3.19.4 Time delay relays**

Provide time delay relays as follows:  
Adjustable over the full timing range with timing repeatability within  $\pm 12.5\%$  of nominal setting.  
Incorporate light emitting diodes indicating energisation states of relays.

#### **7.3.19.5 Phase failure relays**

Provide separate solid-state adjustable phase failure relays which release at the following:  
< 85% of normal voltage (adjustable).  
Single phase failure.  
Reverse phase sequence after an appropriate time delay.

#### **7.3.19.6 Push-buttons**

Provide push buttons as follows:  
Oil-tight, minimum 22mm diameter, or 22 x 22 mm.  
Rated operational current at least 4 A at 240 V a.c.

Emergency stop devices with mechanical latching to AS/NZS 60947.5.5.

Identify functions of each push-button. For latched STOP or EMERGENCY STOP push-buttons, provide label with instructions for releasing latches.

### **7.3.20 Marking and Labelling**

#### **7.3.20.1 General**

Provide labels including control and circuit equipment ratings, functional units, notices for operational and maintenance personnel, incoming and outgoing circuit rating, sizes and origin of supply and kW ratings of motor starters.

#### **7.3.20.2 Labels on assembly exteriors**

Provide Manufacturer's name and details.

Compliance engraved label to be provided on fascia of switchboards which includes information regarding current rating, fault rating, type test certificate, form of segregation, pollution degree, month and year of manufacture, dimension of switchboard and rated voltage.

Provide designation label (assembly name).

Indicate source of electrical supply.

Label controls and fault current limiters, including the following:

Circuit designation for main switches, main controls and submains controls.

Details of consumers mains and submains.

Incoming busbar or cable rating to first tee-off.

Fuse link size.

#### **7.3.20.3 Labels on assembly interiors**

Provide labels for equipment within assemblies. Locate so that it is clear which equipment is referred to, and so that lettering is not obscured by equipment or wiring.

#### **7.3.20.4 Danger, warning and caution notices**

Busbars: If polymer membrane coating is used without further insulation, provide warning notices on the front cover near the main switch or local main switch and on rear covers, indicating that busbars are not insulated.

Fault current limiters: In assembly sections containing fault current limiter fuses provide caution notices fixed next to the fault current limiters, stating that replacement fuse links are to match the installed fuse link ratings, make and characteristics. Provide separate label stating make and fault current limiting fuse ratings.

Externally controlled equipment: To prevent accidental contact with live parts, provide warning notices for equipment on assemblies not isolated by main switch or local main switch.

Stand-by power: Provide warning notices stating that assemblies may be energised from the stand-by supply at any time.

Anti-condensation heaters: To prevent accidental switching off, provide caution notices for anti-condensation heaters.

Insulation and shrouding: For insulation or shrouding requiring removal during normal assembly maintenance, provide danger notices with appropriate wording for replacement of insulation shrouding before re-energising assemblies.

Positioning: Locate notices so that they can be readily seen, next to or, if impracticable, on busbar chamber covers of functional units and behind the front cover of functional units.

Provide circuit identification labels in the cabling chamber of each functional unit, located next to external terminations.

#### **7.3.20.5 Size**

Engrave as follows:

Assembly designation 10mm black letters on white

Main Switches 7mm white letters on red

Main Equipment 7mm black letter on white

Minor Equipment 4mm black letters on white

Essential Services 7mm white letters on red

Warning Notices 7mm and 4mm red letters on white

#### **7.3.20.6 Schedule cards**

General: For general light and power distribution boards, provide schedule cards of minimum size 200 x 150 mm, with typewritten text showing the following as-installed information:

Submain origin & upstream circuit protection device rating

Location of earth electrode (where applicable)

Light and power circuit numbers and current ratings, cable sizes and type and areas supplied.

Mount schedule cards in a holder fixed to the inside of the assembly or cupboard door, next to the distribution circuit switches. Protect with hard plastic transparent covers.

#### **7.3.20.7 Single-line diagrams**

Provide single-line diagrams for custom built assemblies.

Provide non-fading print, at least A3 size. Laminate and enclose in a non-reflective PVC frame and wall mount close to assembly.

#### **7.3.20.8 Marking cables**

Identify the origin of all wiring. Provide durable labels fitted to each core and sheath, permanently marked with numbers, letters or both to suit the connection diagrams.

All final sub circuit cables shall be ferrule labelled to indicate circuit number.

All control cables shall be ferrule labelled at both ends.

All mains and sub main cables shall be labelled at connections to the protection devices.

### 7.3.21 Final Sub-circuit Distribution Boards

Provide final sub-circuit distribution boards as indicated on the drawings. Distribution boards shall be fully assembled by a specialist switchboard supplier, tested and subject to the manufacturer's quality assurance program prior to delivery to site.

IP Rating:

Minimum IP42 for indoor distribution boards in a dedicated, smoke-sealed switchboard cupboard or switchboard room;

Minimum IP56 for indoor distribution boards located in a general room, open area, or path of travel, for containment of smoke hazards;

IP66 for outdoor distribution boards, 2.0mm 316 stainless steel with rainhood cover, padlockable handle, door stay, bottom entry, and plinth where floor mounted.

Distribution boards shall be:

Fitted with flush mount door handles key lockable with 92268 or CL001 key.

provided with internal finger duct cable management on both sides of chassis for cable management;

a minimum of 240mm deep (including door) for indoor distribution boards and a minimum of 300mm deep for outdoor distribution boards;

Provided with sufficient space and DIN rail mounting provisions for all required ancillary devices, including timers, contactors, time clocks, and separate compartment for BMS equipment (where required);

Provided with separate escutcheons, dividing mullions and barriers to separate each metered section of the distribution board and the main switch section;

fitted with hinged escutcheons, with meters and indicating devices fitted flush;

fitted with non-ferrous gland plates;

have lift-off hinged doors with 3-point locking;

fully welded construction;

provided with NC encapsulated 250A or 400A MCB chassis;

provided with dummy pole filler modules held in place by screw onto the chassis rail for all spare poles (clip in pole fillers are not acceptable);

left with minimum of 30% spare pole capacity on completion (i.e. A 60-pole chassis shall be populated to no more than 42 poles on completion);

left with a minimum of 30% spare capacity on gland plates and cable management for future cable installation, taking care to lay out gland plate cable entries to make future cable installation neat and simple;

Provided with typed circuit card indicating: outgoing final sub-circuit cable sizes, circuit breaker rating, circuit breaker number, description of the circuit; origin of submains supplying the DB; size of submain cables; and size of upstream protection device;

Provided with 1 copy of the fully detailed shop drawings printed in A3 colour, and placed in a clear sleeve mounted on the inside of the DB door;

textured orange or grey paint finish for both indoor switchboards or brushed stainless steel for outdoors.

Distribution boards shall comply with the rest of the Switchboards section of this specification in all other respects.

## **7.4 Installation**

Install switchboards level and plumb in the spaces provided.

Fix wall mounted assemblies using a minimum of 4 non-ferrous or galvanised metal fixings.

Mount floor mounted assemblies on purposed built suitable foundation or plinth, generally reinforced in-situ poured concrete.

Secure assembly plinth to foundation using a minimum of 4 masonry anchors.

Provide assembly mounting structure over switchboard entry pits, and finish off with galvanised steel checker plate or Gatic type lid sections complete with lifting provisions.

Bolt transportation sections together to form a rigid assembly. Complete all necessary inter-panel connections.

Install DNSP equipment as required.

Ensure isolation of dissimilar metals to prevent galvanic corrosion in external locations.

For concrete plinth mounted switchboards provide a bituminous impregnated cork gasket between the switchboard base and the concrete plinth. Provide reinforced concrete foundation / plinth, trowel finished a minimum of 100mm above surrounding ground level. Secure switchboard to concrete base using corrosion resistant fixings suitably isolated from the switchboard metalware.

Continue cable support systems to the assembly and complete cable access in a manner which retains the IP rating of the assembly.

Leave spare capacity at cable entries and gland plates for future circuit cable entries.

For all cables provide glands and gland plates.

For final sub circuit cable entries provide suitable vermin proof cable enclosure.

For distribution switchboards exposed to view, provide cable access cover from switchboard to ceiling and/or floor as appropriate to neatly conceal cables and provide a uniform appearance.

Externally mounted assemblies shall have bottom cable entry.

Fully support all cables inside enclosures using cable duct or trays. Secure cables using non-metallic ties. Support and secure cables to within 200mm of terminations.

Ensure earth bonding throughout the assemblies.

Run internal switchboard cabling at right angles through compartment walls and gland off or seal in the same manner as for switchboard cable entries.

## **7.5 Load Balancing**

Measure loads of each phase at each switchboard and balance the load on each phase to within +/-10% of the average peak load. Check load balancing during thermographic testing undertaken during DLP and adjust load balancing as necessary.

## **7.6 Phase Rotation**

Check phase rotation at all switchboards to ensure they all match the site main switchboard.

## **7.7 Testing And Verification**

Carry out all type tests and routine tests for a PTTA in accordance with Section 8 of AS/NZS 3439.1.

Submit test results for verification.

## **7.8 Completion**

Adjust, set and program all adjustable or programmable equipment.

Carry out all required site tests.

Carry out tests for the operating time of each RCD in accordance with AS/NZS 3760, and record the test result in the Operation and Maintenance Manual.

Replace (at no cost) any RCDs which do not meet the requirements of AS/NZS 3760.

Clean all assemblies and remove all waste.

Energise assemblies in accordance with Workplace Health & Safety and company quality assurance procedures.

Ensure all required insulating barriers and housing have been installed.

## **7.9 Maintenance**

### **7.9.1 RCD Testing**

For all switchboards and distribution boards, the NHP Rapid Test facility shall be installed and used for testing of all RCDs and submit report one month prior to the end of the defects liability period.

Carry out tests for the operating time of each RCD in accordance with AS/NZS 3760, and record the test result in the report.

Replace (at no cost) any RCDs which do not meet the requirements of AS/NZS 3760 during the defects liability period.

#### **7.9.2 Thermographic Test Report**

For all switchboards and distribution boards, carry out thermographic scans and submit report 1 month after practical completion, and again one month prior to the end of the defects liability period.

Thermographic scanning shall be conducted under full load conditions. Coordinate with the building operator to ascertain when this is and carry out the scans at this time.

The load currents on each phase for each outgoing functional unit or conductor shall be measured at the time of the scan and logged in the report.

Carry out all corrective actions recommended in the report at no cost.



## **8. ELECTRIC VEHICLE CHARGING**

### **8.1 General**

Supply and install Electric Vehicle (EV) charging equipment as specified on the drawings.

### **8.2 Quality**

#### **8.2.1 Standards**

SAE J1772

SAE J3068

IEC 62196

AS/NZS 3820 – Essential Safety Requirements for Electrical Equipment

AS/NZS 4417.2 Regulatory Compliance Mark for Electrical Equipment

AS/NZS 3000 Wiring Rules – Appendix P

#### **8.2.2 Submissions**

##### **8.2.2.1 Product data**

Submit product data for all system components.

### **8.3 Products**

#### **8.3.1 Warranty**

EV charging equipment shall be provided with a minimum 5 year manufacturer warranty

#### **8.3.2 22kW Three Phase AC Charger – Bollard Mounted Freestanding**

Provide freestanding bollard mounted 22kW three-phase 400V AC Electric Vehicle Supply Equipment (EVSE) as shown on the drawings.

Each EVSE unit shall come complete with the following:

- 22kW IEC 62196 Type 2 sockets
- Automatic-latching protective flaps
- RFID card reader for integration with third party systems (such as access control systems)
- LED/LCD indicator or display to show function/state of charging
- LAN connectivity
- AS 4755 Demand Response enabled
- Open Charge Point Protocol (OCPP) compatible for billing
- Inbuilt class  $\leq 1.0$  CT's and energy meter
- Minimum IP54 rated
- Minimum IK10 impact resistance
- Auxiliary relay input for fire emergency stop

### **8.4 Installation**

#### **8.4.1 Mechanical Installation**

Install the equipment in full accordance with the manufacturer's instructions.

Bollard mounted EVSE's shall be securely mounted to the concrete slab, or hardstand using masonry anchors as noted in the manufacturer's instructions. Use shims to level the unit before tightening the anchor bolts. Once installed, provide a bead of construction adhesive sealant around the base of the plinth where it meets the concrete.

Whether or not they are shown on the drawings, provide underground electrical and communications conduits from distribution switchboards to freestanding EV chargers, and between DC fast charge terminals and remote AC/DC power conversion units, in accordance with manufacturer instructions. Set out and turn up conduits below plinths of units in the locations recommended in the manufacturer instructions.

Provide bollard protection remote to the EVSE to prevent vehicular impact.

#### **8.4.2 Electrical & Communications Installation**

Electrical circuit protection, cable sizes and circuit arrangement for EV charging equipment that is shown on the electrical drawings is nominal only and should be considered as indicative. The exact electrical requirements may differ depending on the EV charging equipment selected by the contractor, and it is to be assumed that the contractor has informed themselves of the specific requirements of their selected equipment during tender and has made all allowances for the requisite electrical works in association, in accordance with the manufacturer's instructions.

Provide residual current device (RCD) protection for all AC EVSE units. RCD's shall be 30mA Type B or as otherwise recommended by the EVSE manufacturer.

Provide equipotential bonding to the conductive case/frame/plinth of all EVSE and DC fast charge equipment in accordance with the equipotential bonding provisions of AS/NZS 3000. Equipotential bonding conductors shall be a minimum of 6mm<sup>2</sup> Cu. The equipotential bonding conductor and the protective earth conductor may be the same conductor provided it is a minimum of 6mm<sup>2</sup> Cu.

Run cabling into the case of EVSE units as per the manufacturer's instructions, using accessories such as cable compression glands to maintain IP ratings and ensure cables are mechanically protected.

Circuit ratings and cable sizes shall be selected in accordance with the selected EVSE manufacturer's recommendations.

Run compatible data & security cabling to EVSE units which are specified as having LAN, OCPP or RFID capability. Run cabling back to the data rack and/or security access control panel/s as required. Test and terminate cables at both ends.

Liaise with client and other trades as required to interface to other systems, including security, fire, IT systems.

### **8.4.3 Fire and Emergency Shut-off**

#### **8.4.3.1 Fire Panel Automatic Shut-off Interface**

Provide an interface between the EVSE(s) and the Fire Detection & Alarm System or 230v Smoke Alarms to automatically shut down all EVSEs in the event of a fire alarm. A relay output from the FDAS panel shall automatically shut off all EVSE units via direct auxiliary input integral to the EVSE. Liaise with the EVSE manufacturer to ensure this provision is included in the EVSE.

#### **8.4.3.2 Remote Emergency Stop Button**

Provide a remote emergency stop button/s which, if pressed, will shut off all EVSEs instantly.

The emergency shut off button shall be green in colour, with a break-glass style mechanism, mounted near the FIP or as required by the FER.

Provide robust printed signage with contrasting lettering of minimum 25mm high text, stating words to the effect of "ELECTRIC VEHICLE CHARGING EMERGENCY STOP". Signage shall be wall mounted at approximately 2000afl.

Provide sufficient Emergency Stop Buttons so that no button is more than 20 metres away from an EVSE.

## **8.5 Completion**

### **8.5.1 Testing**

Test EVSE to ensure it functions as intended and as per the manufacturer's instructions.

Test all emergency shut off systems to ensure they function correctly.

Test all communications and security system interfaces to ensure they function correctly and as intended.

Test load control/demand response functionality to ensure it operates as intended.

### **8.5.2 Clean Up**

Remove all waste and rubbish from site and leave all components in as - new condition.

## **9. ACCESSORIES, APPLIANCES & DIRECT CONNECTIONS**

### **9.1 General**

Supply and install all accessories and appliances including all switched socket outlets, isolators and switches.

Provide direct connections to equipment via local isolators mounted adjacent to the equipment.

### **9.2 Quality**

#### **9.2.1 Submissions**

Provide product data of all proposed accessories and appliances.

#### **9.2.2 Design**

Determine from the Architect the requirements for accessory colours and finishes prior to ordering.

#### **9.2.3 Uniformity**

Provide accessories of the same manufacture for each accessory type.

### **9.3 Products**

#### **9.3.1 Socket Outlets**

Provide single phase 10A rated switched socket outlets, one make and style throughout, flush mounted with separate cover plates, high impact polycarbonate construction.

Double outlet type shall be arranged under the one cover plate.

Provide neon indicators, integral RCDs and the like as specified.

Socket outlets shall be Schneider Iconic Pro Series with window labels

Provide large rocker type, Clipsal Prestige or approved equal, where indicated.

#### **9.3.2 Industrial-type Socket Outlets**

Provide 3, 4 or 5 pin switched socket outlet combinations, polycarbonate enclosure type, surface or flush mounted as specified, IP65 enclosure rating, one make and style throughout, selected from either Clipsal 56 series or NHP ISO ranges. Provide units with integral RCD's as required. For all equipment supplied under the main contract, provide matching plugs to suit the socket outlet type and fit to equipment, including test and tag if required.

#### **9.3.3 Switches**

Provide 10A fluorescent load rated flush mounted rocker type switch plates and mechanism with separate snap on cover plate.

Switches shall be of the same manufacture as the socket outlets.

Provide neon indicators, key operation and the like as specified.

Multigang stainless steel type switch panels shall be flush mounted with finished stainless steel cover plates and engraved switch designations.

Provide large rocker type mechanisms, Clipsal Prestige or approved equal, where shown.

#### **9.3.4 Weatherproof or Protected Accessories**

Weatherproof accessories shall be polycarbonate or metal as specified with minimum protection classification of IP53 (or higher where specified) to AS 60529.

#### **9.3.5 Isolators**

Provide isolators for all direct connected equipment. Isolator rating and type shall be suitable for the equipment rating.

External isolators or isolators in hose-down areas shall be minimum IP65 and made of UV resistant and chemical resistant material.

Isolating switches for domestic-type cooking equipment shall be of a type to match switches and socket outlets and provided in accordance with AS3000, and with engraved mechanisms indicating what they are switching.

#### **9.3.6 Appliances Generally**

Provide appliances as specified, complete with all controls, controllers and connecting cabling.

Confirm appliance loadings and supply requirements prior to ordering and installation.

### **9.4 Installation**

#### **9.4.1 Accessories, Outlets and Isolators**

Install accessories, outlets and isolators using purpose built mounting boxes, plates, brackets and fixings.

Provide earth bonding connections for all metallic mounting components.

Install to mounting heights as shown or advised by the Architect and Superintendent.

Flush wall mount unless shown otherwise.

Confirm final location of all accessories prior to rough-in. Locations shown on drawings are indicative only. Allow for displacement of up to 1 metre from positions indicated.

Segregate terminal zones by physical barriers to limit maximum voltage to 230 V under one cover plate or in one wall box.

Finish all accessories plumb, level and in line.

Ensure accessories mounted in acoustic rated walls are mounted in acoustic-rated wall boxes and are offset by at least 150mm for outlets mounted on opposite sides of the wall.

#### **9.4.2 Appliances and Direct Connected Equipment**

Provide a suitably rated and labelled isolator adjacent to each directly connected appliance or item of electrical equipment.

Final connection between isolator and equipment for free-standing equipment, appliances and machinery shall be via flexible PVC conduit or heavy duty flexible cord. Conduit or cord shall be secured at both ends with secure couplings or glands, and conduit or cord secured with saddles, clamps and the like to avoid excess cable.

Final connection between isolator and equipment for equipment that may be subject to movement or vibration shall be made by appropriately rated flexible cord, heat rated for cooking and heating appliances, and UV rated for external applications. Flex cables shall be secured both ends with compression glands, with intermediate cable clamps/saddles where appropriate to avoid cable damage.

Final connection to building fixed wiring for domestic-type appliances fixed in joinery shall be via heat rated flexible cord and a suitably-rated plug and socket or junction box.

Shorten flex and plug connections to avoid excessive cable lengths.

#### **9.4.3 Ceiling Fans**

Mount ceiling fans at a minimum height of 2400 mm from the floor to underside of blades. Support fans to manufacturers recommendations.

Where multiple fans are controlled from the one location, install a maximum of 3 controllers on the one wall plate.

Where multiple fans are shown to be controlled from a single controller, provide a fan speed controller suitable for operation of multiple fans.

#### **9.4.4 Smoke Alarms and Heat Alarms**

Design, install and commission smoke alarm systems in accordance with AS1670.6.

Interconnect all smoke alarms within a sole occupancy unit.

Install smoke alarms as far as possible from kitchens, bathrooms and laundries whilst maintaining compliance with the Building Code of Australia.

Maintain adequate separation from ceiling fans.

Connect alarms to local lighting circuit. Permanently identify with a prominent label on the electrical switchboard the sub-circuit supplying the smoke and heat alarms.

### **9.5 Labelling**

Provide labelling for all accessories, including SSOs, switches, isolators and direct connections.

Labels shall indicate distribution board number and circuit number of the origin of the circuit the accessory is connected to (E.g. “DB.A CB.1”)

Labelling for non-weatherproof switches and socket outlets shall be typed printed labels in the windows of the Iconic Pro Series removable face plate

Labels for weatherproof accessories, isolators and other electrical accessories and appliances shall be engraved traffolyte with contrasting colours (white background, black lettering), adhesive backed.

Self-adhesive printed ribbon labels are not acceptable and will be rejected.

## 9.6

### **Completion**

Test all appliances and accessories for satisfactory operation.

## **10. LUMINAIRES AND LIGHTING**

### **10.1 General**

Supply and install all luminaires, lamps and lighting controls as scheduled, specified and shown on the drawings.

Luminaire manufacture, model and/or catalogue number shall be as shown on the drawings.

Any unauthorised luminaire substitutions will void the design certification.

### **10.2 Alternative Luminaire Proposals**

Review of any proposed alternative luminaires will be carried out by Ashburner Francis at our standard hourly rates on an hourly rates basis or with an agreed fixed fee (payable by the proponent of the alternative proposal), and will require the provision of the following for each luminaire type proposed to be substituted:

Photometric files (IES/LDT etc);

Technical Data sheets (customised for the exact product proposed);

Test reports for IESNA LM-79, LM-80, TM-21, TM-30;

Declaration of Conformity to AS60598 & EMC compliance signed by supplier;

A working sample in the same specification as being proposed;

A completed *Alternative Lighting Sample Submission Form* (available from Ashburner Francis on request);

Cost saving (per unit) compared to the specified luminaire.

Any proposed alternative luminaire shall, at a minimum, comply with the remainder of this specification, and have all of the same features and specifications of the specified luminaires.

Review of alternative luminaires does not guarantee an approval.

### **10.3 Lighting Design**

The entire lighting design throughout this project shall be designed and certified by a competent person. Provide a Form 15 signed by a person competent in lighting design, referencing BCA Part F and all relevant Australian Standards for lighting design (e.g. AS1680, AS1158, AS4282, AS2560).

Lux plots of the designs of all internal and external areas shall be provided for review and assessment.

Ensure the illumination power density of all areas complies with BCA Part J. Provide calculations for review upon request.

Produce a lighting design that is fit for purpose within all spaces to provide appropriate levels of illumination, uniformity, and a high level of visual comfort and installation aesthetic.



The lighting design shall use Light Loss Maintenance Factors (LLMF) that match the end of serviceable life test results for the selected luminaires, with the addition of dirt accumulation factors as appropriate for the space, to ensure that the lighting design remains compliant with AS1680.1 at the end of its serviceable life.

Task space work plane lighting levels in areas such as classrooms, kitchens, office areas and the like shall have calculated min/avg uniformity ratio of no less than 0.6 unless approved otherwise.

Corridors, circulation spaces, lounge areas and living areas should use “qualitative lighting design” whereby visual comfort is maximised by the use of indirect lighting, wall washing, up lighting, and illuminating architectural or decorative features of the building. Consult with the architect to ensure lighting design is cohesive with the architectural and interior design and highlights feature aspects of the particular spaces.

Utilise wall washing and up lighting where practical to enhance visual comfort. Ensure wall washing and up lighting is even, avoiding “hot spots”. When wall washing, utilise luminaires with specially designed wall wash optics and/or reflectors, to ensure smooth wall washing from floor to ceiling. Follow manufacturer spacing ratios for locating wall washers. Refer to AS1680.1 Section 10.5 for guidance.

External lighting shall be designed to limit glare and spill to below AS4282 levels and to comply with local council by laws and recommendations. Generally use low glare external luminaires.

All task and utility areas such as GLA’s, learning areas, offices, store rooms, commercial kitchens and the like should generally use 4000k colour temperature light sources.

All living areas, bedrooms, dining areas, music rooms, private study areas, residential kitchens, lounges and the like should use 3000k colour temperature light sources.

External lighting should generally use 3000k colour temperatures.

Ensure all luminaires minimise glare to within the acceptable UGR limits for task areas.

Ensure luminaires are of high quality aesthetic design and construction suitable for the spaces.

Ensure all luminaires are designed and selected for ease of maintenance and accessibility appropriate for the location.

## **10.4 Quality**

### **10.4.1 Standards**

All luminaires shall meet the following standard as applicable to that luminaire type:

AS/NZS 60598 series for luminaire construction

ZHAGA Standard for LED interchangeability

IESNA LM-79 for LED Electrical and Photometric Measurement of LED Luminaires

IESNA LM-80 for LED Lumen Maintenance of LED light sources

IESNA TM-21 for projected Lumen Maintenance of LED light source

IESNA TM-30 and CIE CRI (Ra) for colour rendering properties

EMC Compliance to AS/NZS CISPR 15

IEC 62717:2014 LED modules for general lighting - Performance requirements

Lighting design & installation shall comply with the following standards as applicable:

AS1680 series (interior lighting)

AS1158 series (exterior lighting)

AS4282 (obtrusive lighting)

AS2560 series (sports lighting)

#### **10.4.2 Submissions**

Provide samples where requested.

Provide product data sheets and test reports for all luminaires

Provide product data for lighting control systems and dimmers.

Provide luminaire compliance certificates upon request.

### **10.5 Products**

#### **10.5.1 LED Luminaires**

Provide LED luminaires as specified.

Luminaires and/or luminaire components shall have RCM certification and be on the ERAC register.

LED Luminaires shall be purpose-designed for use with LED light sources. Traditional style lamp luminaires fitted with retrofit-style LED lamp units will not be accepted, except where specified (such as feature pendant lights and decorative luminaires).

Illumination of visible diffusers shall be even. LED 'spots' shall not be obviously visible through the diffuser of any opaquely diffused luminaire.

Luminaires shall use optically-designed lenses, reflectors, and/or diffusers to control the output of light and to mitigate glare to below AS1680 UGR levels.

#### **10.5.2 Warranty**

All LED luminaires shall be covered by a minimum 5 year manufacturer repair or replacement warranty. Warranty is to include all components of the luminaire including housing, driver and LED light source.

If any LED luminaire fails or falls below the "end of life" performance levels within 5 years of practical completion, the luminaire shall be repaired or replaced.

Luminaires shall be sourced from manufacturers with local, manufacturer-endorsed agencies for warranty and after-sales services purposes. Luminaires sourced via direct import from manufacturers or distributors without a full-time agent in Australia will not be accepted.

Provide warranty statements for all luminaires and include within the O&M manuals.

#### **10.5.3 LED Light Engines**

Unless specified otherwise, LED luminaires shall incorporate replaceable LED light engines compatible with the Zhaga modular LED specification or an equivalent standard.

Actual spectrally tested Correlated Colour Temperature (CCT) of LED shall be within +/- 100 Kelvins from the specified temperature (e.g. 4000k), and all LED chips shall have a CCT binning quality tolerance of no greater than 3 MacAdam Steps, and must be maintained within this tolerance throughout its rated life.

LED luminaires shall have a Colour Rendering Index (CRI,  $R_a$ ) or TM-30 Fidelity Rating  $R_f$  of no less than 80, or higher where specified.

#### **10.5.4 LED Photometric Performance (IESNA LM79)**

The Complete LED luminaire package (luminaire + driver) shall have published electrical and photometric results from a laboratory or lighting test facility that is a signatory to ILAC (International Laboratory Accreditation Cooperation) such as a NATA laboratory or international equivalent.

All LED luminaires shall have measured photometric performance to support claimed performance in accordance with IESNA LM79.

All photometric files shall be of an “absolute photometry” method.

#### **10.5.5 LED Lumen Maintenance (IESNA LM80 & TM21)**

LED luminaires shall have published IESNA TM21 depreciation projections from a NATA or international equivalent certified lighting test facility which are based on actual test results.

Each luminaire type shall have minimum TM21 projected performance equal to or better than the following characteristics:

L80 B10 @ 50,000 hours @ 25 Degree C for interior luminaires

L80 B10 @ 50,000 hours @ 40 degrees C for exterior luminaires and luminaires used in industrial environments

where “L” = projected lumen depreciation at end of life, and “B” = the overall percentage of luminaires that are projected to drop below “L” at a given time (e.g. 50,000 hours).

E.g. For L80 B10, at 50,000 hours no greater than 10% of the luminaires will have dropped below 80% of their original lumen output.

Where quoted TM21 projections are those of the LED module manufacturer and not that of a complete tested luminaire, the luminaire manufacturer must be able to prove, through in-situ temperature measurement (ISTM), that the thermal management of the luminaire is within the

tolerances specified by the LED module manufacturer in order for the LED module TM21 projections to remain valid for that luminaire in which it is used.

In addition, specified luminaires may also have a spot failure rating, which may or may not be expressed as a third “C” figure. Where a specified luminaire has a manufacturer published spot failure rate, any proposed alternatives shall have equal or lower spot failure rates.

The test conditions and rated characteristics of any specified luminaire overrule these standard minimum requirements. Any proposed alternative must be able to demonstrate equivalent or better performance to the specified luminaire under equivalent or more onerous test conditions.

#### **10.5.6 Luminaire Construction**

Provide proprietary luminaires and associated equipment as specified.

Luminaires shall:

- be thermally designed to ensure optimum operation and component operating temperatures below recommended limits;
- be rigid, true and symmetrical in construction;
- have support and fixing mechanisms to suit the intended application;
- have photometric performance equivalent to the published data;
- be constructed of durable and corrosion resisting materials suitable for the installed conditions and environment;
- be complete with all required lamp holders, terminal blocks, fuses, control gear, temperature rated cabling, diffusers, reflectors, light sources, cable clamps, and fixings as required for complete luminaires;
- have control gear and light sources matched for optimum performance;
- have purpose built low loss electronic controllers for LED lamps;
- have terminal blocks suitable for up to 2 x 2.5mm<sup>2</sup> cables or flex & plug; and
- have Regulatory Compliance Mark (RCM/C-Tick) certification.

Steel sheet metal construction shall:

- be minimum 0.8mm thick zinc-anneal sheet, free from blemishes with all exposed joints welded, filled and rubbed smooth; and
- Be finished using approved polyester powder-coat surface treatment and paint.

Recess mounted luminaires shall be provided with flex and AU 3 pin plug top.

Diffuser material shall be acrylic, polycarbonate or similar UV stabilised plastic product suitable for use as a diffuser in luminaires.

#### **10.5.7 Luminaire Control Gear & Power Supplies**

For non-dimmable LED's provide constant current or constant voltage electronic drivers suitably rated for the load.

For dimming type LED fittings provide purpose built electronic controllers rated for the load, with DALI, DSI, or phase dimming as specified.

DALI dimmable ballasts shall be DALI certified and shall be auto-addressing.

Provide ballasts suitable for the specified luminaire and control gear arrangements, taking into account requirements such as remote mounting of control gear, dimming and lighting control systems.

#### **10.5.8 Power Quality**

Luminaire power factor shall be 0.95 – 0.99 lagging.

Luminaire voltage Total Harmonic Distortion (VTHD) shall be no more than 5% to IEEE standards.

#### **10.6 Controls**

Provide lighting controllers including switches, dimmers, movement sensors, daylight sensors, timers and the like as shown on the drawings and as specified.

Smart lighting controls shall be installed and implemented by the following methods:

- Automated Lighting Control
- Movement Sensor Control
- Daylight Dimming
- Timeswitch control

Dimming shall be solid state and suitable for the load and lamp type.

Dimmer control type shall be DALI as shown on the drawings.

Switch and dimmer controls and scene panels shall be of the type as shown on the drawings.

Provide compatible data and control cable.

#### **10.7 External Luminaire Poles & Fittings**

Provide hot dip galvanised steel poles complete with rag bolt foundation assembly, component hatch secured with vandal-proof bolts, outreach arm, mounting spigot and corrosion resistant fixings.

Foundations shall be of approved structural design for the installation conditions and environment.

All external luminaire poles shall have fused take-offs mounted at the base of the pole with access from the component hatch. Pre-wire fittings between the luminaire and the take-off fuse. Fuse carriers shall be Legrand Modular for industrial cylindrical cartridge fuses or approved equal.

## **10.8 Installation**

### **10.8.1 General**

Install luminaires in complete accordance with manufacturer instructions. Where ambiguity or conflict exists between manufacturer's instructions, and any mandatory Australian standard or the design documents, seek clarification before installing.

Install luminaires in line, parallel or square to the building structure, level, at the same mounting height, plumb and flush with surrounding surfaces.

Support luminaires using proprietary supports, battens, trims, noggings, ceiling roses, chains, suspension rods, trunking, ducting, stainless steel braided wire or lighting track as appropriate.

Fix and secure luminaires to ensure permanent mounting under service conditions. Provide a minimum of 2 fixings per luminaire and more as required to avoid movement or distortion.

Install luminaires so that there are no light leaks.

Mount luminaires and control gear clear of insulation and flammable materials in accordance with AS/NZS 3000 Clause 4.5.2.3.

Size conductors supplying ELV lamps to ensure volt drop is minimised.

Install PE cell switches in locations suitable for reliable performance.

Make wiring connections using flex and plug or hard wiring as appropriate.

Adjust lamp locations and aim luminaires.

Install LED type luminaires to ensure heat dissipation in accordance with manufacturer's requirements.

### **10.8.2 Electronic Components**

Install electronic components including electronic ballasts, LED drivers and LED arrays in locations suitable for limitation of temperature rise to within manufacturer's specified limits.

Follow manufacturer's instructions for the allowable quantity of luminaires on a circuit, ensuring the installation is within the safe limits of the circuit protection for earth leakage and inrush current levels to eliminate nuisance tripping or premature failure of switching relays.

### **10.8.3 Recessed Luminaires**

Co-ordinate requirements with ceiling installer for all cutouts, trims and additional supports for recessed luminaires.

### **10.8.4 Suspended Ceilings**

Support recessed luminaires on ceiling grid or trimming materials as required to carry the weight of the luminaire, in accordance with AS/NZS 3000 Clause 4.5.2.4.

#### **10.8.5 High Bay / Low Bay Light Fittings**

Support high bay or low bay pendant suspended luminaires to manufacturer's recommendation. In addition, provide stranded stainless steel safety wire with crimped eyelet fixings.

#### **10.8.6 Lighting Controllers**

Install all lighting control systems and run cabling for control and power supplies as required.

Arrange controls in groups or zones as shown on the drawings.

Carry out all required programming and data installation.

Provide automatic control of artificial lighting for energy management purposes in accordance with Clause J6.3 of the Building Code of Australia.

#### **10.8.7 External Lighting**

Secure to structure using corrosion resistant fixings.

Maintain IP rating at cable entry and fixings. Use circular cables for entering all exterior light fittings through compression glands or rubber grommets. Flat TPS cable is not acceptable for exterior light fittings as it will not create an effective seal with the proprietary glands and grommets found in exterior light fittings.

Install blockouts in masonry or in-situ concrete for recessed luminaires.

Provide cable entries to manufacturers recommendations to maintain weather or water proof enclosure.

Install cabling between components to ensure moisture protection integrity is maintained. Run continuous cable lengths between components.

Provide in-situ concrete footings to structural design requirements for all post or pole mounted luminaires, bollards and in-ground uplights. Exposed concrete shall be trowel finished and gaps between concrete foundation and base plates shall be grouted and finished smooth.

Aim all external lighting systems as required after sunset.

#### **10.8.8 Wiring to External Lighting**

Install double insulated cabling for all external lighting circuits in accordance with Specification Section 5 "Cables and Cable Supports".

#### **10.9 Completion**

Test and verify operation of all luminaires.

Program all lighting control systems.

Adjust all controls in conjunction with the Superintendent for system operation as required.

Clean all luminaires and diffusers to as-new condition.

Provide all necessary access equipment for luminaire and lighting controls commissioning



## **11. EMERGENCY EVACUATION LIGHTING**

### **11.1 General**

Supply and install emergency evacuation lighting complying with the requirements of the National Construction Code - Building Code of Australia.

The system shall be centrally monitored and able to be interrogated externally via the BAC communications network

### **11.2 Quality**

#### **11.2.1 Standards**

- AS/NZS 2293.1 for system design, installation and operation.
- AS/NZS 2293.2 for inspection and maintenance
- AS/NZS 2293.3 for construction of luminaires.

#### **11.2.2 Technical Data**

Submit technical data and AS/NZS 2293.1 classification data for each emergency and exit luminaire type.

Submit details of proposed central testing and monitoring system showing system topology and details.

#### **11.2.3 Design**

Ensure system compliance with the Building Code of Australia.

### **11.3 Products**

#### **11.3.1 Emergency Luminaires**

Provide emergency luminaires manufactured generally in accordance with Specification Section "Luminaires and Lighting".

Batteries for emergency luminaires shall be lithium iron phosphate type rated at a minimum life of 7 years at 25°.

Mark each battery with date of installation using an indelible pen.

Emergency Luminaires shall be covered by a minimum 5 year manufacturer warranty, with separate battery warranty.

### **11.4 Installation**

Install emergency luminaires in accordance with Specification Section 10 "Luminaires and Lighting" and the requirements of AS/NZS 2293 Series.

Provide a 230V unswitched active supply to each emergency luminaire for mains failure sensing in each control area.

Arrange lighting controls so that emergency lights are activated on the failure of normal lighting in the vicinity. Ensure that failure of any lighting control circuit that causes failure of the normal lighting in an area also activates the emergency lighting in that area.

Provide a log book or other provision for the recording of maintenance in accordance with the requirements of the AS/NZS 2293 Series.

Label each emergency luminaire with a unique identifier and record in the log book or central monitoring system database.

## **11.5 Completion**

Prior to testing connect mains supply for at least 24 hours to enable full battery charge.

Carry out tests under both simulated mains failure and test switching conditions to ensure satisfactory system operation.

Replace faulty components and retest the affected zone.

Confirm battery charging function after completion of test.

Fill out the log book or populate the database with the details of all emergency lights installed in accordance with the requirements of the AS/NZS 2293 Series.

Provide Queensland Building Act Form 16 Certification to confirm correct installation of the emergency lighting system.

Carry out 6 monthly procedures in accordance with AS/NZS 2293.2 just prior to practical completion and twice during the defects liability period. Record all results in log book/database

## **12. BUILDING LIGHTNING PROTECTION**

### **12.1 General**

Supply and install a building lightning protection system if determined during design. Perform an analysis to AS1768 to determine if a LPS is required.

Surge filtering for copper cables entering and/or exiting the building are specified in separate sections of this specification.

### **12.2 Quality**

#### **12.2.1 Standards**

- AS/NZS 1768 for lightning protection system.
- AS/NZS 3000 for equipotential bonding.

#### **12.2.2 Submissions**

Provide product data for all components of the proposed system.

Provide details of proposed means of minimising galvanic corrosion, connections to building structure and facade, expansion joints and earth pit arrangement.

#### **12.2.3 Design**

Confirm compliance with AS/NZS 1768

### **12.3 Products**

#### **12.3.1 Air Terminals, Bonding Conductors and Down Conductors**

Supply purpose built rods, rectangular section, saddles and items a necessary for a complete, bonded building lightning protection system.

Components shall typically be copper or may be another material which complies with AS/NZS 1768 and is more suitable for the building structure and finishes.

Down conductors shall be separate external type or the structural steel as detailed on the drawings.

Earth pits shall allow sufficient space for earth electrode bond and testing.

Earth electrodes shall be copper clad steel, minimum 1800mm long.

### **12.4 Installation**

Install building lightning protection air termination and down conductor components as follows:

- Positioned and secured to complement building finishes and aesthetics.

- Secured by fixings and brackets which do not compromise the structural integrity or waterproofing qualities of the building components.
- Fitted with expansion loops or joints for long horizontal or vertical runs.

Bond all roof structure, roof sheeting, projecting metal such as flues, antennas, stays, support structures, roof mounted equipment and gutters.

Steel reinforcing bars used as down conductors shall have 1 bar per column welded to form a continuous length with Cad welded threaded connectors at ends of the columns.

Exposed down conductors shall be bonded to the building façade as required.

Bond all incoming metal piped services to the lightning protection system to the requirements of AS/NZS 1768.

Install driven earth electrodes or buried earth electrode grid as required.

Carry out site earth resistivity tests to determine required earth termination system.

Located driven earth electrodes in dedicated pits labelled to indicate function.

Provide drilled earths complete with resistivity enhancer where site conditions require this installation method.

## **12.5 Completion**

Carry out tests in accordance with AS/NZS 1768.

## **13. FLOODLIGHTING POLES**

### **13.1 General**

Supply and install high mast lighting poles as shown on the drawings, complete with engineered footings suitable for the prevailing conditions. Poles shall be supplied and installed by a specialist high mast light pole manufacturer, with in-house local engineering capability to design, manufacture, install and certify poles and footings fit for purpose for the location and function as intended by the electrical lighting design.

### **13.2 Standards**

The pole is to be continuously tapered, hot dip galvanised and multi-sided, conforming with the requirements of all relevant standards, in particular:

AS/NZS 4677:2000 Steel utility poles

AS/NZS 4676:2000 Structural design requirements for utility services poles

AS/NZS 1170.0:2002 Structural design actions Part 0: General principles

AS/NZS 1170.1:2002 Structural design actions Part 1: Permanent, imposed and other actions

AS/NZS 1170.2:2011 Structural design actions Part 2: Wind actions

AS4100 – 1998 Steel structures

AS/NZS 4600:2005 Cold-formed steel structures

AS/NZS 1554.1:2014 Structural steel welding – Welding of steel structures

AS1275-1985 Metric screw threads for fasteners

AS/NZS 1112.1:2015 ISO metric hexagon nuts - Style 1 - Product grades A and B

AS/NZS 1112.2:2015 ISO metric hexagon nuts - Style 2 - Product grades A and B

AS 3600-2009 Concrete structures

AS/NZS 4680:2006 Hot-dip galvanized (zinc) coatings on fabricated ferrous articles

AS 1798-2014 Lighting poles and bracket arms – Preferred dimensions

British Standard BS EN 40-3-3:2013 Lighting columns - Part 3-3: Design and verification - Verification by calculation

AS 1657-2013 Fixed platforms, walkways, stairways and ladders-Design construction and installation

AS/NZS 1252:1996 High strength steel bolts with associated nuts and washers for structural engineering

### **13.3 Importance Level**

AS/NZS 1170.0:2002 Structural design actions Part 0: General principles and the Building Code of Australia (the BCA) (refer Tables B1.2a and 1.2b) - Poles shall be designed for Importance Level 1 for poles less than 20m and Importance Level 2 for poles 20m and over. Design working life shall be a minimum of 50 years.

Importance Level 1 has a design wind event with a 1:100 annual probability of exceedance in non-cyclonic areas.

Importance Level 2 has a design wind event with a 1:500 annual probability of exceedance in non-cyclonic areas.

#### **13.4 Wind Loading**

Wind loading on the column shall be determined using the relevant annual probability of exceedance obtained from the NCC/BCA.

The fraction of critical damping shall be taken as 0.05 (ultimate) and 0.01 (serviceability) for poles with more than two- (2) overlaps and 0.02 (ultimate) and 0.005 (serviceability) for all others. The natural frequency of the pole shall be calculated considering varying diameters and thicknesses over the height of the pole and using a 1.1 safety factor for the mass at the top of the pole. Poles are typically wind sensitive structures with natural frequencies of less than 1. A second order analysis shall be performed.

#### **13.5 Deflection**

Pole deflection at serviceability wind speeds shall be less than 6.7% for poles under 30.0m and less than 3.3% for poles 30.0m and over or for poles with integral access, i.e. climbing rungs.

#### **13.6 Welding**

Welds shall be by a continuous automatic gas shielded electric arc process complying with the relevant parts of AS1554 Structural steel welding. The longitudinal seam welds on pole sections shall conform to GP standards while baseplate and spigot welds shall conform to SP standards as mentioned in AS1554. Weld sizes shall be verified by a qualified structural engineer and specified in the engineering report and on workshop drawings.

#### **13.7 Foundation Bolts**

All poles are to be baseplate mounted and founded on a cast-in-situ concrete footing. A galvanised foundation bolt assembly complete with positioning template and two nuts and washers per bolt is to be provided to suit the pole baseplate. Foundation bolts must be manufactured from deformed reinforcing bars with a nominal yield stress of 500 MPa. Bolts shall be threaded in accordance with AS1275-1985 Metric screw threads for fasteners and fitted with class 5 nuts in accordance with AS/NZS 1112.1:2015 ISO metric hexagon nuts Style 1. Foundation bolts shall be tied to a suitable reinforcing cage. The length of each foundation bolt must allow for the length of the thread above ground, a minimum of 100mm cover and a development length in accordance with AS3600-2009 Concrete structures, Section 13. The underside of the baseplate shall be grouted in accordance with Clause 24.16 in this Section and to the Superintendent's satisfaction. Grout performs a structural function in the design of the pole and stability of the pole may be undermined if it is neglected.

#### **13.8 Access Doors & Gear Trays**

Flush fitting access doors shall be provided to house luminaire control equipment. The pole section at the access door shall be reinforced and analysed according to British Standard BS EN 40-3-3:2013 Lighting columns - Part 3-3: Design and Verification - Verification by Calculation, or

a recognised finite element analysis package. Poles housing more than eight sets of control gear shall have a vented bottom access door.

Refer to BAC guidelines on the positioning of the access doors with respect to trafficable areas and luminaires above.

Access doors shall be fixed to the pole using a minimum of two standard hex key bolts.

Provide gear trays behind access doors for mounting of lighting control gear, electrical switchgear and associated electrical accessories. Gear trays shall be manufactured as part of the pole prior to hot dip galvanising to ensure that the integrity of the galvanising is maintained. Where possible, threaded studs or pre-drilled bolt holes shall be provided for mounting of equipment to gear trays. All holes drilled in the gear tray for mounting equipment shall be deburred and treated with a rust-resisting paint.

Provide a screw-fixed, all-weather, UV resistant engraved label to each access door with wording similar to:

WARNING!

400 VOLTS  
LIVE PARTS BEHIND

ACCESS BY QUALIFIED PERSONNEL ONLY

“WARNING” lettering to be WHITE, minimum 10mm high  
Other lettering to be WHITE, minimum 5mm high  
Background to be RED

### **13.9 Design Calculations**

The section capacity of the pole shall be analysed over a minimum of 100 increments according to AS4100 – 1998 Steel Structures and AS/NZS 4600:2005 Cold-formed steel structures. Pole manufacturers shall submit a fully documented computer printout verifying structural adequacy and conformance to current Australian Standards and this specification. Printouts shall include all input parameters including drag coefficients, cross-sectional properties and design assumptions authorised by a qualified structural engineer and include a statement of compliance with all aspects of this specification. Calculations shall prove pole section capacity, access door reinforcing and foundation bolt suitability.

### **13.10 Luminaire Maintenance Provisions**

#### **13.10.1 External Luminaire Maintenance**

Luminaires shall be accessed by external machinery e.g.- cherry pickers. Luminaire cross-arms shall be designed to ensure easy access of fittings for re-lamping and aiming. Climbing rungs and maintenance platforms are not required.

### **13.11 Accessories**

All cross-arms shall be supplied with structural bolts (certified to AS1252 and tested as per the Australian Steel Institute Tech Note #1-06) for mounting luminaires, and cross-arms shall be suitable for aiming luminaires in the required directions without shadowing or clashing occurring.

### **13.12 Corrosion Protection**

The pole and all steel accessories shall be hot dip galvanized after fabrication in accordance with AS/NZS 4680:2006 Hot-dip galvanized (zinc) coatings on fabricated ferrous articles.

### **13.13 Tolerances**

Manufacturing tolerances shall be in accordance with AS 1798-2014 Lighting poles and bracket arms – Preferred dimensions. In particular the poles shall be checked for compliance with the straightness requirements of this standard: 0.3% of height, as detailed in Clause 5.1.

### **13.14 Assembly and Erection**

Assembly and erection shall be carried out only by qualified rigging personnel. Assembly and erection instructions are to be provided with every pole over 6.0m mounting-height.

### **13.15 Grouting Procedure**

The underside of all pole baseplates shall be grouted with a non-shrink grout with a minimum strength of 32MPa at 28 days within seven days of installing the pole.

Grouting the underside of pole baseplates serves the following functions:

Grouting plays a significant part in the structural capacity of the pole and foundation bolts in resisting high wind forces.

Preventing injury to persons through electrocution caused by handling or tampering with exposed cables;

Preventing damage to electrical cables by vermin;

A Ø16 weep hole shall be installed under the baseplate to the exterior to facilitate condensation from inside the pole to drain to the surrounding ground.

### **13.16 Certification**

For work designed for and carried out in Queensland, all geotechnical, civil and structural engineering work associated with the design of poles and footings shall be carried out in accordance with the Professionals Engineers Act (QLD) and certified by competent Registered Professional Engineer of Queensland (RPEQ). Provide copies of the RPEQ certification.